WHAT DO I DO WITH THIS LARGE INFORMED CONSENT PACKAGE?

1) **READ EVERY PAGE IN THE NEXT 2 WEEKS OR LESS.**
2) **INITIAL EVERY PAGE. YOU CAN INITIAL WITH ONE LETTER—YOUR FIRST OR LAST. For example, write "John Smith = J," and put a J on every page.**
3) **SIGN EACH PLACE IT SAYS TO SIGN AND DATE IT OR IT IS OF NO USE.**
4) **FAX TO 239 304 1987 IF YOU HAVE A VERY RELIABLE QUALITY FAX MACHINE OR MAIL OVERNIGHT OR TWO DAY OVERNIGHT TO:**

JAMES SCHALLER, MD, MAR
14555 INDIGO LAKES CIRCLE
NAPLES, FL 34119

WE REGRET THAT ANTI-PHYSICIAN AND ANTI-MEDICAL FREEDOM LEGAL EVENTS IN THE USA REQUIRE THIS IMMENSE LEVEL OF INFORMED CONSENT.

BABESIA, ATYPICAL BARTONELLA, LYME AND OTHER MEDICAL CONSENTS

MOST PHYSICIANS DO NOT AGREE WITH HIS APPROACH

Dr. Schaller will *not* treat you in any area of medicine with a diagnosis or a treatment that traditional physicians, traditional medical societies or traditional insurance company reviewers might reject, unless he has your full informed consent. You need to be sure you feel fully informed. If not, Dr. Schaller will give you more information on various options, including information from those that feel his approaches are poor.

Dr. Schaller's positions are often not consistent with traditional medical societies, traditional infection societies, traditional OB/GYN societies, traditional endocrine societies, traditional aging and preventative disease societies.

The only approved treatments in the USA are typically surgery or large pharmaceutical company medications. Routine allopathic

medicine does not tend to embrace other treatment models. Dr. Schaller feels that treatment options and dosing sometimes need to be broader or altered.

<u>If you want the routine standard of care in all areas of medicine, you are working with the wrong MD</u>. If you only want medications used in FDA approved manners at FDA approved dosing levels and durations, you are working with the wrong MD. Dr. Schaller recognizes that most medications are used in off-label manners, and his dosing is based on his personal clinical experience —he rejects all guidelines and protocols as simplistic and untailored for individual problems. He is also dealing with issues that he feels many medical societies are behind on. They would say he does not have the solid huge published studies to support some of his ideas. They would be right.

<u>SIGN_____ DATE_____</u>
<u>PRINT NAME_____</u>

Dr. Schaller Does Not Feel 30 People From an Infection Society, Using Possibly Filtered Studies on Chronic Lyme Disease, and Possibly Ignoring Other Studies, Should Dictate to 800,000 MD's, Who Are Treating 300 Million Unique Americans, How to Treat Their Tick-Borne and/or Flea-Borne Infections.

Antibiotic treatment for Lyme is too risky

I was dismayed by your July 12 story on Lyme disease, not only because I was the lead author of the Infectious Diseases Society of America's guidelines on the disease, which the article criticized, but also because I, like many doctors, have felt the frustration of seeing a patient with unexplained illness for whom I can offer no effective medical treatment.

Too often, patients with mysterious symptoms are misdiagnosed with Lyme disease. Often they are prescribed prolonged antibiotic treatment, an unnecessary and potentially harmful approach. Antibiotics have risks as well as benefits. Controlled clinical trials have shown no benefit from prolonged antibiotic therapy for treating Lyme disease. It causes unnecessary patient discomfort, inconvenience and

expense and may actually be harmful. It also can promote drug-resistant infections in the community.

Although patients sometimes report feeling better after taking a drug, that does not necessarily prove cause and effect. In one clinical trial of long-term treatment for Lyme disease, almost 40 percent of patients given a placebo felt much better. Rigorous scientific studies are essential.

IDSA's guidelines were based on an extensive review of the scientific evidence. In our view, the risk of long-term antibiotic treatment for Lyme disease outweighs the theoretical benefit.

Dr. Gary P. Wormser
Infectious Diseases Society of America
Valhalla, N.Y.

SIGN_____ DATE_____
PRINT_____

<u>*LYME TREATMENT*</u>
<u>INFORMED CONSENT AND RELEASE</u>

During the last ten years Dr. Schaller has become involved in many areas of medicine outside his original training.

These areas have developed because patients were not getting well, and Dr. Schaller wanted to find solutions.

Specifically, Dr. Schaller and much of his family are from a county with a high incidence of Lyme in SE Pennsylvania, Further, many of his relatives have contracted Lyme and other deer tick infections. Many developed other infections carried by ticks and rarely tested for in traditional medicine.

Originally, he studied Lyme and sought out mentors, merely to learn to order the best labs, so if people were positive they could be referred to their family physician for treatment. Yet over the years, Dr. Schaller has developed the **<u>minority medical opinion</u>**, that most national labs are poor at detecting the presence of Lyme. He feels most lab testing done by massive national labs is poor at picking up Lyme in *obviously positive* patients. He feels this applies to all three of the common Lyme tests—the Lyme titer, the Western Blot and a PCR. Most physicians trust these tests would disagree with Dr. Schaller on his lack of faith in these large national labs.

Further, when Dr. Schaller referred patients with subtle or serious Lyme symptoms to infectious disease physicians, the patients were upset with the indifference and inability of these doctors to listen and treat their symptoms. They were "blown

off" according to these patients. They only improved briefly, at best, with their treatments.

Finally, Dr. Schaller came across some progressive mentors who were also annoyed at the poor labs used for testing and the sloppy approach to diagnosis, e.g., no investigation into co-infections, areas of vacations, and other relatives with the illness. He was introduced to other lab options, and found that these *dedicated* Lyme, Babesia or Bartonella labs, e.g., IgeneX, Fry Clinical Labs, and special indirect tests done at large labs had results that *fit the clinical complaints and history offered.*

So Dr. Schaller found that he was able to diagnose better. However, when he moved to Florida and as he continued to aggressively learn, he realized the best physicians to treat this illness were in other states or Columbia Medical Center, and his patients were not willing or able to travel so far for treatment.

So he decided, after receiving some ongoing modest instruction, and doing aggressive self-study into thousands of articles, to begin to treat individuals with about 4 tick borne illnesses.

However, you must understand some basic things fully.

First, Dr. Schaller makes *no claims* to be an infection expert, a Lyme expert, an antibiotic expert or to be following labs and protocols used by most physicians in any specialty. If you want "standard" CDC treatment you should not use Dr. Schaller to treat you. If you want another opinion, please eagerly seek it.

If you can find someone else to treat your Lyme, we encourage you to use him or her.

Bluntly, Dr. Schaller has *no desire* to build a Lyme practice. While he likes educating on this topic, he has many other research interests and other ways to serve you.

He merely offers to treat those who have a certain belief they have no other functional option.

Lyme frequency, preferred labs, and treatments are very polar topics in some communities. Dr. Schaller has no desire to be a part of that battle. He has no compulsion to win you over or convince you about your need to think as he thinks. Dr. Schaller's position on Lyme is complex and involved, and he sees this knowledge base changing monthly. But currently his Lyme, Babesia and Bartonella beliefs are clearly the *minority* medical opinion.

All medications, including antibiotics, have risks of liver and/or kidney damage and rarely a risk of death. The risks are far too many to fully list. Always read at least 3 articles on the risks and side effects of any medication you take. *The doses used to treat your Lyme are higher than common doses used among most physicians.* Never assume any dose is a common dose. Assume it is considered a dose or a combination of antibiotics that are considered too high or fully unnecessary by infection MD's.

Dr. Schaller does no physical exams, and defers these to your *required internist or family physician*, who must approve of your care.

Finally, to be a patient of Dr. Schaller's he must have some contact with you every 3 months, even if you are fully healthy. *But if you have Lyme*, he must have contact with you 3x a month for the first 2 months and then no less than every other week. Not all these contacts must be in person, but they cannot be merely by email or fax. He simply will not practice sloppy medical care with a possibly *serious and complex illness.*

I FULLY UNDERSTAND THIS MATERIAL ABOVE AND WOULD LIKE DR. SCHALLER TO TREAT MY CHILD OR MYSELF.

**SIGN _____ PRINT _____DATE_____

FDA STAFF MEMBER THINKS MANY LYME PHYSICIANS ARE VERY BAD

I believe in allowing patients to read as much negative material as possible about my positions. I do not care if it is true, but only that some people in various medical societies, government agencies and other physicians say it. Here is a nice collection of common criticisms of physicians who feel Lyme is a common health issue that often requires more than 21-30 days of antibiotics. Most infection experts in the USA feel Lyme is easy to cure and is easily diagnosed with virtually any routine lab. I have watched this position fail in the care of many relatives, friends, physician patients and vocal patients who speak for themselves on this matter. Nevertheless, I love offering the words of those who despise my beliefs.

Here are the criticisms of an FDA employee. I appreciate her simple candor.

FDA EMPLOYEE CRITICISMS

1) Since Lyme disease is difficult to diagnose and the treatments are still not completely agreed upon in the medical community, this provides

opportunities for unscrupulous practitioners.

2) In my view, there is no reason to put patients (especially children) on months, or sometimes years, of antibiotic treatments, when there is no conclusive scientific proof that they have Lyme disease.

3) The fact [is] that over-use of antibiotics is extremely harmful.

4) I don't think these practitioners are being honest with their patients about the bad side effects of these [antibiotic] treatments.

5) Also, many of these practitioners send their patients blood samples to one particular diagnostic lab in California, and coincidentally all of the samples turn out to be positive for Lyme disease. [This is obviously IGeneX Labs at IGeneX.com and I will let them address this point. Their number is 800 832 3200. Their Medicare and CLIA lab license is on this web site.]

6) I am concerned about the so-called 'epidemic' of Lyme disease diagnosis in this country

7) [I am concerned]… these experts are charging patients $500 - $1,000 per visit, and they do not participate in health insurance plans.

8) By the way, do you think there are any poor people out there who have Lyme Disease, and would these Lyme experts even be willing to see them without up front payment?

9) Would you be willing to trust your life to practitioners who have been involved in misconduct, who have legal defense funds, and who falsify information on their resumes? I think not. Where there is smoke, there is fire.

_____**********_____

We appreciate her simple and clear criticisms of those who have tried to offer useful Lyme disease care.
If you want another position on her criticisms, log on to the Lyme Disease Association web site at: **http://www.lymediseaseassociation.org** or the International Lyme and Associated Diseases Society (ILADS) web site at: **http://www.ilads.org**

ILADS is a small group of multi disciplinary physicians who disagree with most infectious disease experts in the US, and feel Lyme labs are often poor, and that treatment of chronic Lyme should not be cut off at 30 days.
Some think medical truth is best understood after hearing opponents speak, and perhaps also reflecting on your own treatment experience. In America this is the foundation of our justice system—opponents present their arguments and they are weighed in the balance. Others feel the truth is what most specialists think. If you want the position of most infection specialists, you do not want to see an ILADS physician and you would use your local lab and accept treatment of Lyme that is limited to a month.

_____

I do not fear giving folks the most insulting characterization of physicians who are working hard to treat Babesia, Chronic Lyme and Atypical Bartonella.

Sign_____ Date_____Print

Name_____

BABESIA, BARTONELLA, EHRLICHIA AND MYCOPLASMA INFORMED CONSENT

Dr. Schaller has written three books on Babesia and also did one on Bartonella. He has read hundreds and perhaps 1-2,000 articles about Babesia and Bartonella. Yet, he certainly does NOT feel like he is an expert in any of these infections. He does feel Bartonella is not rare. He does feel that Ticks that carry Lyme often also carry Babesia and most infection experts feel this makes someone sicker, i.e., having Lyme plus other infections such as Babesia.

1) **DR. SCHALLER MAKES NO CLAIM TO KNOW *ANYTHING* ABOUT ANY OF THESE INFECTIONS.**

2) **HE DOES NOT CLAIM TO BE THE BEST IN YOUR LOCATION TO TREAT YOU FOR THESE INFECTIONS.**

3) ***HE WOULD BE DELIGHTED TO TRANSFER YOUR CARE TO ONE OF THREE INFECTION SPECIALISTS WHO YOU MIGHT CONSULT* ON THESE INFECTIONS. CONSIDER CALLING YOUR LOCAL HOSPITAL OR COUNTY MEDICAL SOCIETY TO GET THESE OTHER EXPERTS FOR SECOND, THIRD AND FOURTH OPINIONS.**

4) **DR SCHALLER WILL ONLY TREAT YOU IF YOU REFUSE TO HAVE OTHERS TREAT YOU, YOU CAN FIND NO OTHER OPTIONS, AND *YOU UNDERSTAND HIS APPROACH IS 100% DIFFERENT THEN THE APPROACH OF INFECTION SPECIALISTS*.**

5) **YOU UNDERSTAND ALL TREATMENTS HAVE LIVER, HEART, BRAIN, AND KIDNEY RISKS AND CAN KILL.**

6) **In many cases, Babesia presence in blood may be very low and infected patients may have negative smears. This is especially likely in chronic infections in patients**

who still have their spleen. (Maryland Department of Health and Mental Hygiene (DHMH) **http://www.dhmh.state.md.us/labs/pdf/critlink/aug06critilnk.pdf**

7) **SIGN _____ DATE_____ Understand the above? YES/NO**

Persistent Lyme Disease Consent

There is considerable uncertainty regarding the diagnosis and treatment of Lyme disease. No single diagnostic and treatment program for Lyme disease is universally successful or accepted. Medical opinion is divided, and two schools of thought regarding diagnosis and treatment exist. Each of the two standards of care is described in peer-reviewed, evidence-based treatment guidelines. Until we know more, patients must weigh the risks and benefits of treatment in consultation with their doctor.

Your Diagnosis. The diagnosis of Lyme disease is primarily a clinical determination made by your doctor based on your exposure to ticks, your report of symptoms, and your doctor's observation of signs of the disease, with diagnostic tests playing a supportive role.

Doctors differ in how they diagnose Lyme disease.

- Some physicians rely on the narrow surveillance case criteria of the CDC for clinical diagnosis even though the CDC cautions against this approach. These physicians will fail to diagnose some patients who actually have Lyme disease.

For these patients, treatment will either not occur or will be delayed.

- Other physicians use broader clinical criteria for diagnosing Lyme disease. These physicians believe it is better to err on the side of treatment because of the serious consequences of failing to treat active Lyme disease. These physicians sometimes use the antibiotic responsiveness of a patient to assist in their diagnosis.

Your doctor also may need to rule out other possible causes of your symptoms, such as arthritis, lupus, syphilis, Reiter's syndrome, MS, CFS, and FMS. If you are unclear of your diagnosis, be sure to discuss this with your doctor.

Your Treatment Choices. The medical community is divided regarding the best approach for treating persistent Lyme disease. Some physicians think that the long-term effects of Lyme disease are caused by damage to the immune system and are therefore unaffected by antibiotics. Others believe that the infection persists, is difficult to eradicate, and therefore requires long-term treatment with intravenous, intramuscular, or oral antibiotics, frequently in high and/or combination doses.

Your treatment options include:

1. **Not pursuing antibiotic treatment and, if appropriate, seeking symptomatic relief for your continuing symptoms;**

2. **Treating your illness with antibiotics until clinical resolution of your symptoms, regardless of duration of treatment; or**
3. **Treating your illness with antibiotics for thirty days only.**

If you elect to pursue antibiotic treatment, you will be treated with antibiotics selected to address the Lyme bacteria as well as any other tick-borne co-infections you may have, such as Ehrlichiosis, Babesiosis, or Bartonellosis. Your doctor will sometimes recommend IV medications when there is neurological involvement, carditis, complicated Lyme arthritis, or inadequate response to oral medications. Sometimes treatment consists of IV antibiotic treatment followed by oral antibiotics. Other classes of drugs may be needed to treat non-bacterial tick-borne diseases such as babesiosis.

Potential Benefits of Treatment:

Antibiotic treatment may result in improvement in your clinical condition. Although there is substantial evidence that most patients improve with continued treatment, not all patients improve with treatment (See Attachment A).

Patient response varies widely:

- Some patients experience substantial improvement of their symptoms and do not require further treatment,
- Some patients feel worse initially during treatment, before improving,
- Some patients improve with antibiotic treatment, but relapse when treatment is stopped, and
- Some patients do not respond to antibiotic treatment. Sometimes persistent symptoms represent permanent changes to a patient's body, in which case further antibiotic treatments may be of no further benefit. Other times the illness progresses but, for unknown reasons, does not respond to additional treatment.

Risks of treatment: There are potential risks involved in using any treatment. Some of the problems with antibiotics can include allergic reactions, which may manifest as rashes, swelling, and difficulty with breathing. These problems may require medications to reverse the allergic reaction and may require emergency treatments. Other potential complications include stomach and bowel upset, abdominal pain, diarrhea, or bowel irritation, which may require interruption of the antibiotic and prescribing other medications to manage the digestive upset. It is also possible that secondary infections such as yeast infection of the skin, mouth, intestinal, and

genital tracts may occur in some people, causing discomfort and the need for corrective therapies. Some physicians worry about resistant super infections with antibiotics over 30 days. It is also possible that other problems such as adverse effects on liver, kidneys or other organs may occur. For oral antibiotics, one source believes it is estimated that the risk of major side effects is 1 in 10,000 and the risk of minor side effects is 4 in 100.[1]

Factors to consider in your decision: The Infection Society of America and virtually all Medical Associations feel they know **the exact duration of treatment for all patients for treatment of Lyme**. It is about a month of treatment. A small minority of physicians feels no one knows the optimal treatment of symptoms that persist after being diagnosed with Lyme disease and treated with a simple short course of antibiotic therapy. The appropriate treatment may be supportive therapy without the administration of any additional antibiotics. Or, the appropriate treatment might be additional antibiotic therapy. If additional antibiotic therapy is warranted, no one knows for certain exactly how long to give the additional therapy. By taking antibiotics for longer periods of time, you place yourself at greater risk of developing the side effects such as a C. Diff. infection that

can harm your intestines. A minority of physicians feel by stopping antibiotic treatment, you place yourself at greater risk that a potentially serious infection will progress. Antibiotics are the only form of treatment shown to be effective for treating Lyme disease, but not all patients respond to antibiotic therapy. There is no currently available diagnostic test that can demonstrate the eradication of the Lyme bacteria from your body. Other forms of treatment designed to strengthen your immune system also may be important. Some forms of treatment are only intended to make you more comfortable by relieving your symptoms and do not address any underlying infection.

Your decision about continued treatment may depend on a number of factors and the importance of these factors to you. Some of these factors are listed below:

- The severity of your illness and degree to which it impairs your quality of life
- Whether you have co-infections, which can complicate treatment
- Whether you have been responsive to antibiotics in the past
- Whether you relapse or your illness progresses when you stop taking antibiotics

- Your ability to tolerate antibiotic treatment and the risk of major and minor side effects associated with the treatment
- Your willingness to accept the risk that, left untreated, a bacterial infection potentially may get worse

For example, if your illness is severe, significantly affects the quality of your life, and you have been responsive to antibiotic treatment in the past, you may wish to continue your treatment. However, if you are willing to accept the risk that the infection may progress or if you are not responsive to antibiotics, you may wish to terminate treatment. Be sure to ask your doctor if you need any more information to make this decision.

Based on this information, I have decided: (CHECK ONE)

☐ To treat my Lyme disease with antibiotics <u>until my **clinical** symptoms resolve</u>.

☐ Only to treat my Lyme disease with antibiotics for **<u>thirty days</u>**, even

☐ I do not want to pursue antibiotic therapy

I want to think this over each month with Dr.

if I still have symptoms.

Schaller and make a tailored decision that I like for me. I reject a set cook book formula

To my knowledge, I am not allergic to any antibiotics except those listed below, and I will ALWAYS remind Dr. Schaller when we are talking about ANY antibiotic:

_____
_____
__

Signature:

_____

Date: _____

Print Name:

_____

Attachment A. Lyme Disease Treatment Studies

No clinical studies have determined the optimal antibiotics or the optimal duration of treatment. Controlled clinical studies to date have been limited and conflicting, although two out of the three studies showed improvement on retreatment. Non-controlled studies also indicate that most patients improve with continued treatment. The controlled and non-controlled studies are described below.

Controlled studies of Persistent Lyme Disease			
Study	**Treatment**	**Results**	**Comments**
Krupp (2003)[2]	4 weeks of IV ceftriaxone	64% showed improvement on fatigue No improvement on cognition	Cognition finding criticized because subjects were not selected based on cognitive impairment. Hence, improvement on this scale would not be expected since no initial impairment was demonstrated.
Klempner (2001) [3]	4 weeks IV ceftriaxone then 2 months of oral doxycycline	No improvement on SF-36.	Study criticized because subjects were not selected based on fatigue. Improvement on this scale would not be expected since no initial impairment was demonstrated. ILADS has issued a position paper outlining in detail substantial criticisms of the study.[4]
Fallon (2004) [5]	10 weeks of IV Ceftriaxone	Cognitive improvement	Preliminary results as data are still being analyzed and official study is being written.

Despite the current focus on controlled studies, some researchers note that there is a high correlation

between controlled and observational studies and that they “usually produce the same results”.[5] In addition, non-controlled studies often provide more clinically-relevant treatment information because they deal with the diversity of patients seen in practice and allow for more flexibility in terms of treatment approach.

Non-Controlled Studies Supporting Longer Treatment Approaches or Retreatment	
Study	**Comments**
Oksi (1999) [6]	9 of 13 patients (69%) with disseminated Lyme disease who were initially treated for 3 months with oral or IV antibiotics but subsequently relapsed had good response to retreatment with IV ceftriaxone for 4-6 weeks. “ [T]reatment… with appropriate antibiotics for even more than 3 months may not always eradicate the spirochete and long term antibiotics may be necessary.”
Donta (1997) [7]	277 patients with chronic Lyme treated for between 1 and 11 months: 20% were cured, 70% improved and 10% had treatment failure.
Oksi (1998) [8]	30 patients with disseminated Lyme treated for 100 days, 90% with good or excellent responses: “prolonged courses of antibiotics may be beneficial in this setting”.
Wahlberg (1994) [9]	Success rates for 100 patients with late Lyme disease: 31% (4 of 13) with 14 days of ceftriaxone; 89% (50 of 56) with ceftriaxone, then 100 days of amoxicillin and probenecid; and 83% (19 of 23) with ceftriaxone, then 100 days of cephadroxil.
Fallon (1999) [10]	18 patients retreated either with intravenous, intramuscular or oral antibiotics scored better on overall and individual measures of cognition. Those retreated with intravenous therapy showed greatest improvement.

The studies described above are necessarily limited to the choice of antibiotics tested and the duration that the antibiotics were given in the study. They do not tell us what would happen if patients were

treated with different antibiotics or for longer periods of time.

SIGN_____ DATE_____

PRINT_____

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4. The International Lyme and Associated Diseases Society (ILADS), *Evaluation of antibiotic treatment in patients with persistent symptoms of Lyme disease. April 2003. http://www.ilads.org/position2.html*
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11. Wickham, R., S. Purl, and D. Welker, *Long Term Central Venous Catheters: Issues for Care.* Seminars in Oncology Nursing, 1996. **8**(2 (May)): p. 133-147.

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SIGN_____ DATE_____

PRINT NAME_____

RISK OF PROLONGED ANTIBIOTIC USE

The largest infectious disease society in the US feels Lyme is fully treated in 30 days. This generally is felt to include severe and chronic cases. Therefore, if at any time you want to stop your antibiotics at ANY TIME, feel free to do so. I am willing to treat individuals who feel certain they want longer treatment and it is the best option for them, but only after you have read at least 5 sources of information that ***oppose long treatment with antibiotics for over 30 days.*** **We request strongly you read three articles about any antibiotic you use. Look closely at the side effects and interactions. The Internet offers many useful sites such as:**

1) http://www.drugs.com/
2) http://www.rxlist.com/
3) http://www.mayoclinic.com/health/drug-information/DrugHerbIndex
4) http://www.nlm.nih.gov/medlineplus/druginformation.html

******Make Sure You Learn FULLY About Any Antibiotic You Take.**

****SIGN_____**
****PRINT_____ DATE_____**

Dr. Schaller does not claim to be an expert in anything. In keeping with Florida law, he is merely saying he is very interested in the infections carried by ticks and fleas, and makes no claims as being smarter or more concerned than other physicians.

****

Be Careful of self-proclaimed medical experts

Your feature article on Lyme disease (Monday June, 14) contained many undocumented statements which demand a response.

The article - is based on a report by the International Lyme and Associated Diseases Society (ILADS), a group which represents a decidedly unconventional and minority position regarding Lyme Disease. **Judging from their literature, most of the physicians associated with this group appear to be self-designated Lyme disease experts, but not actually infectious disease specialists. None of their board of directors or officers lists any academic positions at major institutions** recognizing their expertise in infectious diseases and immunology. In contrast to positions stated in

your article , most recognized specialists in their field believe that 70-80 percent of patients with acute Lyme disease have a history of a classic round or oval rash known as ECM. The ELISA test, contrary again to the statements of the ILADS group, is a highly sensitive test for Lyme disease, although it may take four to six weeks to become positive. Furthermore, Lyme disease in the acute stage is highly curable (greater than 95 percent) with a single 14-21 day course of oral doxycycline or amoxicillin. These facts are amply documented in the medical literature.

In 24 years of family medicine in Chester County I have seen several hundred cases of acute Lyme disease. I cannot recall a single failure after a standard course of oral antibiotics.

When Lyme disease is not diagnosed in its acute state, a variety of symptoms, including joint swelling and on occasion, cardiac and neurologic manifestations can develop. Everyone agrees that these cases can be more difficult to treat. However, even in late-stage Lyme disease, a month-long course of oral antibiotics or a few weeks of intravenous antibiotics are usually curative.

. As a family physician who is comfortable with incorporating alternative and complementary aspects in to my medical practice, I recognize that the medical establishment is not always correct. An open exchange of ideas is a vital component of scientific inquiry and it is good that alternative views are expressed on medical issues. However, truth and integrity require that when a group of physicians makes statements that appear to contradict the overwhelming weight of scientific evidence, they should inform the public that their opinions are clearly unconventional. Judging from their literature, members of this group certainly believe fervently that they are right and are critical of scientific studies which seem to contradict their position. In fact, they seem to believe so strongly in their position that I wonder if this colors their objectivity.

Almost all alternative or complementary approaches to medicine involve using few toxic drugs and attempting to heal illness through non-pharmacological means, such as nutritional supplements, herbal remedies, acupuncture, massage, etc. What is alarming about the recommendations of the ILADS group is that they advocate high doses of antibiotics for a very extended period of time, far in excess of what is generally prescribed. With everything we know about the dangers of overusing antibiotics, including the recently discovered correlation between antibiotic use and breast cancer in

women, I feel that the recommendations of the ILADS group run the risk of doing considerable harm to patients who follow its guidelines.

Increasing numbers of patients nowadays suffer from vague but chronically debilitating symptoms including fatigue, depression, and muscle and joint pains. Some of these patients are diagnosed with Chronic Fatigue Syndrome or fybromyalgia, but no one really knows what causes these diseases and conventional medicine is often unable to help these people. So they frequently go from one practitioner to another seeking a diagnosis and an elusive cure. Many of these patients will spend months or years pursuing worthless, expensive and potentially dangerous treatments for Lyme disease without a shred of evidence that they ever had it.

The co-signatories of this letter represent leading experts in infectious disease and Lyme disease in the southeastern Pennsylvania region. At the very least, your readers need to know that we do not agree with many statements made by the ILADS group. By not consulting with the acknowledged experts in the field, the writer of your article has presented a decidedly one-sided view of Lyme disease and has done your readers a disservice.

RICHARD G. FRIED, M.D.
Board-Certified, Family Practice
Kimberton Clinic

STEVEN J. WEBER, M.D.
Board-Certified, Internal Medicine and Infectious Diseases; Infectious Disease Consultant, PMMC, Phoenexville and Grandview Hospitals

STEPHEN J. GLUCKMAN, M.D. F.A.C.P.
Professor of Medicine, University of Pennsylvania, Director, Infectious Diseases Clinical Services, PENN Care
Board-Certified, Internal Medicine and Infectious Diseases

I HAVE READ OR WILL READ PROMPTLY THIS CAUTION LETTER ABOVE

SIGN_____ DATE_____
PRINT_____

BARTONELLA THOUGHTS

DR. SCHALLER HAS STUDIED BARTONELLA FOR YEARS. HE FEELS IT IS A SERIOUS INFECTION IN MANY PEOPLE AND NOT A MERE COLD OR SIMPLY JUST "CAT SCRATCH FEVER," BUT THE TOP ARTICLES ON IT, REPORT OVER 20 TYPES OF DAMAGE TO ALL MAMMAL ORGANS.

THE BELIEF WE ARE EXPERTS IN THIS GLOBAL AND EXPLODING INFECTION, WHICH IS PASSED MANY WAYS TO HUMANS, SEEMS NIAVE.

Bartonellosis:
Clinical and diagnostic challenges

Apr 1, 2010
By: **Edward B. Breitschwerdt, DVM, DACVIM**
VETERINARY MEDICINE

Prior to 1990, only two pathogenic *Bartonella* species, *Bartonella bacilliformis* and *Bartonella quintana*, were known to infect people, whereas bartonellosis was not yet known as a disease in pet or domestic animals. **Since 1990, more than 22 *Bartonella* species have been described, of which at least half have been implicated or confirmed as animal or human pathogens**. [!!!]

When a genus of bacteria is discovered or, in the case of *Bartonella*, rediscovered, numerous clinical, microbiological, and pathological concepts related to disease causation and microbial pathogenesis are sequentially redefined. Subsequently, the medical relevance of the genus undergoes continued maturation as knowledge of the organism, the host immune response, diagnostic test sensitivity and specificity, treatment efficacy, and epidemiology expand. Since the early 1990s, this paradigm of discovery, rediscovery, and ongoing biological and medical redefinition has clearly been applicable to the genus *Bartonella*. This rapidly expanding body of clinically relevant information has been summarized in three recent reviews.[1-3]

A UNIQUE PATHOGEN

On an evolutionary basis, these highly fastidious, gram-negative bacteria have become adapted to one or more mammalian reservoir hosts (*e.g.* cattle, deer, elk, sheep, cats, squirrels, groundhogs, kangaroos, dogs, rats, rabbits), within which *Bartonella* species usually cause a long-lasting endotheliotropic infection, accompanied by a relapsing intraerythrocytic bacteremia. Also, an increasing number of arthropods, including biting flies, fleas, keds, lice, sand flies, and ticks, are confirmed or suspected vectors for *Bartonella* species transmission among various animal populations.

In the context of intravascular pathogens, *Bartonella* species may prove to be unique, as these bacteria can induce intraerythrocytic

infection and vascular endothelial infection and, experimentally, can achieve intracellular localization within monocytes, macrophages, dendritic cells, and CD 34+ progenitor cells. In the reservoir-adapted host and perhaps in the accidentally infected patient, ***Bartonella*** **bacteremia may be chronic, potentially spanning months to years in duration. Although incompletely studied, these bacteria also appear to induce immune suppression through a variety of mechanisms. Collectively, these factors appear to contribute to a diverse spectrum of disease manifestations in people and dogs, including arthritis, endocarditis, epistaxis, granulomatous inflammation, lymphadenopathy, meningoencephalitis, osteomyelitis, panniculitis, and vasoproliferative lesions**.[1-5]

A DIAGNOSTIC CHALLENGE

For the diagnostician, confirming infection with a *Bartonella* species remains extremely challenging. Bacterial isolation, serology, and PCR amplification of *Bartonella* species DNA directly from patient samples each have substantial diagnostic limitations. Enrichment culture of blood, cerebrospinal fluid, joint fluid, or various pathological effusions by using *Bartonella* alpha Proteobacteria Growth Medium (BAPGM—Galaxy Diagnostics) before PCR is performed is often required to document active infection with a *Bartonella* species.[2,4,6,7]

Studies from our laboratory have shown that about half of infected dogs will not have detectable antibodies to *Bartonella* species test antigens.[2,6,8] Although unproven, it appears that chronic intravascular infection with a *Bartonella* species may induce a degree of immunological anergy, resulting in undetectable levels of organism-specific antibodies in some chronically infected patients.

A CONTINUING PURSUIT

Although numerous and important advances have increased our understanding of bartonellosis, there is a substantial need to better characterize various modes of transmission, the duration of infection, deleterious effects on immune function, cellular pathogenesis, optimal treatment regimens, and risk factors for zoonotic *Bartonella* species infections. Our research to date indicates that veterinary professionals are at risk for occupational exposure to numerous *Bartonella* species carried by pet and wild animals.[2,9,10] In the context of public health, veterinarians should continue to eliminate and control arthropod vectors on animals; avoid bites, scratches, and contact with animal saliva; and exercise caution when obtaining and handling blood and other diagnostic samples.

Editors' note: Dr. Breitschwerdt is the chief scientific officer for Galaxy Diagnostics, which provides the BAPGM test.

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Bartonellosis: An emerging and potentially hidden epidemic?

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Emerging Diseases
Patients at the crossroads of new diseases and chronic ills.
by **Pamela Weintraub**



Pamela Weintraub is a senior editor at Discover magazine and author of *Cure Unknown: Inside the Lyme Epidemic.*

Babesia: The malaria-like disease in your yard, part 1

Malaria mimic in the U.S.
Published on April 4, 2009

It was sometime after the New Year when, feeling better but hardly well after initial treatment for Lyme disease, that I drove a mile down my hill for a latte at Starbucks and my appointment with my Lyme practitioner. She greeted me with a smile, waving a paper in my face. She seemed so pleased she was literally aglow.

A few weeks before, dissatisfied with a plateau in my treatment response, my returning migraines and fatigue, she'd drawn blood and sent it to Quest Lab for a babesia test. "When Lyme disease patients don't get well," she told me, "coinfection with babesiosis is often the cause." …

The pieces were falling into place. With my spiking headache and continued exhaustion, babesiosis certainly made sense for me. A decade earlier, in 1990, I'd spent a couple of weeks as a science-writing fellow at the Marine Biological Laboratory at Woods Hole, Massachusetts, right across the water from Nantucket lsland, where human babesiosis had been studied more than fifteen years before.

It was during the early seventies, before Lyme was even recognized, that Andrew Spielman, a tropical medicine expert at the Harvard School of Public Health, was asked to investigate what locals called Nantucket fever. At the time, only two patients were known. The first was a wealthy Nantucket woman who came down with a disabling mystery illness marked by extreme anemia, fatigue, and fever that

local doctors could not explain. So she chartered a plane to Rutgers, New Jersey.

Rutgers doctors examined her blood under a microscope, diagnosed her with malaria, and placed her on the standard treatment, chloroquine. When the treatment didn't work, they grew alarmed because drug resistant malaria is, after all, a threat to public health. A slide of blood was shipped off to the CDC, where experts identified not malaria but another similar agent that also inhabits red blood cells-*Babesia microti*, known to cause cattle epidemics that wiped out entire herds. With the identification of her infection, the woman was finally treated correctly and got well.

When a second case of babesiosis appeared on Nantucket Island a few years later, physicians again were stymied. But the second patient happened to be friends with the first, and finally, with doctors throwing up their hands, it fell to the first patient to diagnose the disease in the second. Her lay diagnosis was correct, and the second patient was treated and recovered as well.

That's when Spielman entered the fray. Would he care to find the cause of these cases in the environment? Observing the cycles of infection year after year, he finally tracked babesia through the ecosystem, discovering that it lived in the blood of mice and spread from one mammal to the next through the bite of an *Ixodes* tick. Larva and nymphal (baby and adolescent) ticks ate by sucking the blood of mice and other small mammals like shrews and chipmunks. But adult ticks, far bigger in size, generally required larger mammals like deer for a blood meal. It was only in geographic areas with an abundance of large mammals that *Ixodes* ticks could mature to adulthood and reproduce substantially enough for the disease to spread.

When the spirochete *Borrelia burgdorferi* was identified by Willy Burgdorfer as the cause of Lyme disease in 1981, Spielman realized that the newly discovered illness involved the same tick and same natural cycle he'd already charted for human babesiosis on Nantucket Island and beyond. For Spielman, the connection between the two infections-Lyme and babesiosis-was immediately clear. Every year since his first discoveries on Nantucket, after all, new babesia patients had been diagnosed.

And some of them had exhibited not just the malarialike headache and fever spikes typical of babesia, but also a confounding circular red rash and strange pains that migrated from joint to joint-symptoms classically associated not with babesiosis, but rather Lyme disease. "There was a woman who lived opposite our field station who had migratory arthritis and the expanding rash and-babesiosis. She obviously had Lyme as well, but no one had the organism, and restrictive case definitions came into play," Spielman recalled, "because in the beginning, the differential diagnosis for Lyme disease included a travel history to Lyme, Connecticut!"

Eventually the boundaries for both infections expanded at different but proportional rates, and with each infection requiring a treatment ineffective for the other, the coinfected patients of Nantucket Island provided important clues to the spectrum of disease. Sometimes physicians just needed to treat babesiosis for an intractably sick "Lyme" patient to get well. Even as babesiosis extended its range and came to rival Lyme disease as a cause of illness, the lessons of Nantucket's coinfected patients would fall on deaf ears.

Indeed, few Lyme disease patients were ever tested for, or had even heard of, babesiosis; and though the two epidemics had been spawned in tandem from the start and could be equally debilitating, few primary care doctors in endemic areas ran the babesiosis test. "We don't test for that," our Westchester county pediatrician explained at the time.

The internist who tried to treat my headaches-classic for babesiosis-never mentioned the possibility that infection, either Lyme or babesiosis, might be a cause.

Yet in retrospect I believe the Babesia diagnosis was my missing link.
…

Pamela Weintraub is a Senior Editor at Discover Magazine. The post above is adapted from her book on Lyme disease and its coinfections, entitled ***Cure Unknown: Inside the Lyme Epidemic****, published in 2008 by St. Martin's press.*

AT A GLANCE

- The most common tick-borne illness, Lyme disease, is treated with a short course of antibiotics.
- Treatment guidelines, developed by the IDSA, were investigated by the Connecticut Attorney General.
- After an extensive voluntary review process, a special independent panel unanimously concurred that IDSA's 2006 Lyme disease guidelines are medically and scientifically valid and don't need to be changed.
- The Review Panel agreed with the guidelines, which stated that long-term antibiotic treatment is unwarranted and potentially dangerous.

SPECIAL REVIEW PANEL UNANIMOUSLY UPHOLDS LYME DISEASE TREATMENT GUIDELINES

Short-term Antibiotics Proven to be Best Treatment for Patients

[April 22, 2010, ARLINGTON, Va.] – A special Review Panel has unanimously agreed that no changes need be made to the 2006 Lyme disease treatment guidelines developed by the Infectious Diseases Society of America (IDSA), the nation's pre-eminent authority on infectious diseases.

The guidelines, voluntarily used by physicians to help them decide what treatments are best and safest for patients, were the subject of an antitrust investigation begun in 2006 by Connecticut Attorney General Richard Blumenthal. Blumenthal examined whether the guidelines' authors had conflicts of interest and failed to consider divergent medical opinions. IDSA maintains those assertions were unfounded and, as part of an agreement to end the Attorney General's investigation, voluntarily agreed to a one-time special review of the guidelines by an independent scientific Review Panel whose members were certified to be free from any conflicts of interest by an independent ombudsman.

Lyme disease is caused by an infection with a type of bacterium called *Borrelia burgdorferi*, which typically is transmitted by the deer tick. Lyme disease often causes a red, circular rash and muscle and joint aches. A short-course of oral antibiotics remains the standard treatment for patients with Lyme disease, the most common tick-borne illness, which first was discovered in Lyme, Conn.

At issue is the existence of a "chronic" form of Lyme disease and its treatment with long-term antibiotics for months or years. This controversial and costly treatment, which is typically delivered intravenously through a catheter, is advocated by self-described "Lyme literate" physicians.

"IDSA's primary concern is for the health and safety of patients," said IDSA President Richard Whitley, MD. "Our goal is to ensure that patients are given treatment that is safe, effective and supported by scientific evidence. Allegations that the 2006 guidelines panelists stood to gain from the recommendations are unfounded. In fact, one could argue the opposite was true, because the panelists recommended short courses of generic drugs."

To ensure a fair and unbiased review of the 2006 guidelines, IDSA and the Attorney General jointly selected an ombudsman, Howard Brody, MD, PhD, an author and respected medical ethicist at the University of Texas Medical Branch. Dr. Brody screened all potential Review Panel members for conflicts of interest using criteria jointly approved by the Attorney General and IDSA. The Review Panel was comprised of nine scientists and physicians from relevant disciplines who had not served on any previous Lyme disease guidelines panel.

The Review Panel's work took more than a year to complete, with the group convening 16 times, including at an all-day public hearing in Washington, D.C. More than 150 individuals or organizations submitted evidence and other information including letters, newspaper articles, patient medical records and other materials on Lyme disease and its treatment. **Learn more about Lyme disease and the review process**.

"This was truly an extraordinary review process," said Carol J. Baker, MD, who chaired the Review Panel and is a pediatric infectious diseases specialist at Baylor College of Medicine, Houston. "The final report includes more than 1000 citations, and after thoughtfully and carefully reviewing myriad materials, the panelists feel confident that advice to physicians treating Lyme disease should remain the same as set forth in the 2006 Lyme disease guidelines."

The Review Panel concurred that all of the recommendations from the 2006 guidelines are medically and scientifically justified in light of the evidence and information provided, including the recommendations that are most contentious: **that there is no convincing evidence for the existence of chronic Lyme infection; and that long-term antibiotic treatment of "chronic Lyme disease" is unproven and unwarranted. Inappropriate use of antibiotics (especially given intravenously) has been shown to lead to deadly blood infections, serious drug reactions and *C. difficile* diarrhea, as well as the creation of antibiotic-resistant bacteria or "superbugs."**

As noted in the **Review Panel report**, "In the case of Lyme disease, there has yet to be a single high quality clinical study that demonstrates comparable benefit to prolonging antibiotic therapy beyond one month." The report also noted that "the inherent risks of long-term antibiotic therapy were not justified by clinical benefit."

At the request of the Attorney General, the Panel also reviewed a statement in the guidelines' executive summary concerning diagnostic testing. The Panel was evenly split on whether the statement would benefit from modification or clarification. In its final report, the Panel offered this guidance to clinicians: "Symptoms that are commonly attributed to chronic or persistent Lyme, such as arthralgias, fatigue, and cognitive dysfunction, are seen in many other clinical conditions and are, in fact, common in the general population…. It would thus be clinically imprudent to make the diagnosis of Lyme disease using these nonspecific findings alone." On the other hand, **the Panel said, "in clinical practice, the presence of certain classic complications of Lyme disease such as aseptic meningitis, AV nodal block,**

inflammatory arthritis, and cranial or peripheral neuropathies, in a patient with epidemiologic risk of Lyme disease and in whom alternative diagnoses have been excluded or are unlikely, may be sufficiently convincing as to constitute an exception to the statement in the Executive Summary."

Dr. Whitley notes that, as per the Review Panel's recommendation and IDSA's practice, IDSA will review the Lyme disease guidelines on a regular basis to determine the need for updating based on any newly available evidence that would warrant a change to the current recommendations. Guidelines involve synthesizing the highest-quality research available to guide physicians seeking to provide the best and safest care for their patients. In addition to Lyme disease, the Society authors treatment guidelines for more than 50 infectious diseases and conditions, including HIV/AIDS, influenza, pneumonia, and *C. difficile*.

"This special review validates the recommendations, which are again proven to be based on the best science available," said Dr. Whitley. "Our number one concern is the patients we treat, and we're glad patients and their physicians now have additional reassurance that the guidelines are medically sound."

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The Infectious Diseases Society of America (IDSA) is an organization of physicians, scientists, and other health care professionals dedicated to promoting health through excellence in infectious diseases research, education, patient care, prevention, and public health. The Society, which has more than 9,000 members, was founded in 1963 and is based in Arlington, Va. For more information, see ***www.idsociety.org****.*

ARE GUIDELINES GOD?

Standards, Practice Guidelines, and Statements Developed and/or Endorsed by IDSA

About Practice Guidelines-Final Except

It is important to realize that guidelines cannot always account for individual variation among patients. They are not intended to supplant physician judgment with respect to particular patients or special clinical situations. IDSA considers adherence to the guidelines listed below to be voluntary, with the ultimate determination regarding their application to be made by the physician in the light of each patient's individual circumstances.

GUIDELINE TROUBLES

This is not a new problem. **Clinical variation** - the way doctors differ in what they do - has been observed for decades. Sir Liam cites the examples of tonsillectomy and hysterectomy as examples - operations which are still performed too often, on too many patients, unnecessarily. Yet angioplasty, the non-surgical procedure for expanding blocked coronary arteries in sufferers from heart disease, is still denied to too many patients for whom it is necessary.

Enormous research effort has been expended on trying to understand the reasons why doctors who know what they should do then go and do something different. **It is not just a lack of knowledge, it is about priorities, custom and practice. But, Sir Liam said, the solution has not been found.**

WHY DO MOST GUIDELINE OR PROTOCOL MAKERS HIDE? DO THEY KNOW THEY ARE SOMETIMES SUED

AFTER HARM COMES FROM FOLLOWING THEIR EDICTS? WHY ARE THEY INTENSELY AND REPEATEDLY REVISED AFTER A MERE PERIOD OF YEARS?

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ONE PHYSICIAN SUGGESTS LONGER ANTIBIOTICS AT TIMES- HOWEVER THE CDC AND THE INFECTION SOCIETY OF AMERICA WOULD REJECT HER POSITION

Late Stage Lyme Disease: Arguments for an Individualized Approach

Deborah A. Metzger, PhD, MD

The purpose of this review is to provide arguments in favor of a more liberal approach to the treatment of late stage Lyme disease, specifically the use of intravenous antibiotics for extended periods of time.

Lyme disease is a chronic, persisting, multi-systemic infection, which is caused by *Borrelia burgdorferi* spirochetes that are transmitted by common deer ticks (Ixodes). Like syphilis, which is another spirochetal infection, Lyme disease may affect several organ systems and proceed through several stages. It may also persist if it is not properly diagnosed and treated in the earliest stage. During the first stage, a pathognomonic bull's eye rash may develop that establishes the diagnosis. It is often accompanied by a flu-like illness. Unfortunately, in 20-50% of those infected with Borrelia, no rash develops, develops in an uncharacteristic form, or is not noticed[1,2]. Without appropriate antibiotic treatment, the disease becomes disseminated resulting in episodic or persistent neurologic, musculoskeletal, or cardiac symptoms. Several lines of evidence suggest the Lyme disease is very much underreported[3] and that perhaps as many as 90% of those affected are not diagnosed.

Ticks that carry the Lyme bacteria also carry co-infections such as Ehrlichia, Babesia, and Bartonella. Approximately 2/3 of patients with Lyme disease have at least one of these co-infections[4] but patients are not routinely tested for them. Patients who have Lyme disease together with a co-infection may remain mysteriously ill and unresponsive to standard treatment. Thus, Lyme is a complex illness potentially consisting of multiple tick-derived co-infections.

Most physicians agree that when treated very early in the course of the disease that most Lyme patients will get well. Also generally agreed is that Lyme disease patients who have gone undiagnosed and now suffer late stage disease may continue to experience debilitating symptoms following a month-long course of antibiotics. All agree that these symptoms-arthritic, neurologic, and mulisystemic-can last for months or years. The most controversial aspect of the treatment of late stage Lyme disease is the optimal antibiotic regimen.

For the vast majority of bacterial infections, a defined course of antibiotics either eliminates the bacteria or decreases the number of bacteria so that the immune system can eradicate the survivors. Lyme disease is not a typical bacteria in that it shares some of the characteristics of more challenging bacterial infections such as mycobacteria and syphilis: it is difficult to routinely culture, has a slow growth rate, can remain dormant for lengthy periods[5], can invade intracellular sites[6,7,8,9], and may sequester in areas where antibiotic

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[1] Jacobs, RA. Infectious Diseases: Spirochetal. Current Medical Diagnosis and Treatment (Lange Medical Books/McGraw-Hill, NY, 39th edition, 2000

[2] Donta ST. Tetracycline therapy for chronic Lyme disease. Clin Infect Dis 1997; Suppl 1:S52-S56

[3] United States of America Department of Health and Hyman Services Food and Drug Administration Center for Biologics Evaluation and Research Vaccines and Related Biological products Advisory Committee Meeting, May 26, 1998

[4] Benach JL, Coleman JL, Habicht GS: Serologic evidence for simultaneous occurrences of Lyme disease and babesiosis. J Infect Dis 144:473–477, 1981

[5] Preac-Mursic V, et al. Formation and Cultivation of *Borrelia burgdorferi* Spheroblast-L Form Variants. Infection 24(1996);3.

penetration is problematic such as the CNS, joint cartilage, and anterior chamber of the eye[10]. To make matters worse, there are no tests that reliably determine when Borrelia has been effectively eradicated. As clinicians, we are left to use our best medical judgment in individualizing care for our patients.

Oral antibiotics are preferred because of the ease of administration and low cost. Intravenous antibiotics are used for infections that are resistant to orally administered antibiotics, when inadequate blood levels are achieved by the oral route, or when penetration into privileged sites (i.e., the CNS) or poorly vascularized tissue (i.e., cartilage) is needed.

Review of the medical literature to determine an evidenced based approach for the treatment of late Lyme disease reveals a paucity of data. The studies that are often quoted as supporting a particular evidenced-based approach to late Lyme disease are summarized in **Appendix A**.

There are several themes that run through these diverse studies:

1. Antibiotics are accepted as mandatory in active Lyme disease treatment. However, the ideal antibiotics, their dosage, route of administration and duration of therapy have not been established.

2. Many patients remain well after a single course of oral, IM or IV antibiotics. However, many other patients with Lyme disease, initially improve while on antibiotics but relapse when antibiotic treatment is discontinued. There is often relief of symptoms when antibiotics are reinitiated, implying persistence of the bacterial infection.

3. Many of the antibiotics used in the studies do not penetrate the CNS, such as doxycycline[11]. Thus, persistence of neuroborreliosis would be expected.

4. The most effective treatments for late Lyme disease include at least 2 weeks of intravenous ceftriaxone or cefataxime. Retreatment protocols, for relapses and treatment failures, include significantly longer treatment courses, i.e., greater than 4 weeks.

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[6] Klempner et al. Invasion of Human Skin Fibroblasts by the Lyme Disease Spirochete, *Borrelia burgdorferi*. J Infect Diseases 1993;67:1074-81.

[7] Ma et al. Intracellular localization of *Borrelia burgdorferi* within human endothelial cells. Infect Immun 1991;59:671-78.

[8] Duray PH, Johnson RC. The histopathology of experimentally infected hamsters with the Lyme disease spirochete, *Borrelia burgdorferi*. Proc Soc Exp Biol Med. 1986: 263-269.

[9] Georgilis et al. Fibroblasts Protect the Lyme Spirochete, *Borrelia burgdorferi*, from Ceftriaxone in vitro. J. Infec. Dis. 1992:166:440-4.

[10] Jaruratanasirkul et al. Distribution of Azithromycin into brain tissue, cerebrospinal fluid, and aqueous humor of the eye. Anti Agents Chemo 40:825-26, 1996

[11] Chopra et al. Tetracycline antibiotics: mode of action, applications, molecular biology, and epidemiology of bacterial resistance. Microbiol Mol Biol Rev 65: 232-60, 2001

5. None of the studies included evaluation and treatment for the co-infections such as Rocky Mountain spotted fever, bartonella, babesiosis, or ehrlichia that are present in as many as 2/3 of patients. This may explain the poor response to treatment in some of the studies using 30 days of IV antibiotics.

6. Persistence[12] of symptoms or relapse is quite common[13,14] implicating that duration of treatment and/or the type of antibiotic used is inadequate. Relapse and failure to respond to intensive antibiotic treatment has been attributed solely to an autoimmune reaction related to the presence of Borrelia[15]. However, there are studies documenting the persistence of *Borrelia burgdorferi* in antibiotic-treated patients[16] and following up to 12 months of intravenous antibiotic therapy[17,18,19]. **Appendix B** summarizes the studies that demonstrate persistence of *Borrelia burgdorferi* after antibiotic treatment.

7. Most of the studies involved highly selected patient populations. Lyme patients present with a broad spectrum of symptoms and response to antibiotics. Thus, the relevance of the conclusions of these studies to most patients with late Lyme disease is problematic.

8. Many antibiotic regimens do not take into account that many antibiotics only kill actively dividing organisms. The fact that some cultures of *Borrelia burgdorferi* have taken up to 10 months to grow suggests that most treatment guidelines recommend a too short period of antibiotic treatment[20]

9. Given the range of symptoms related to Lyme disease and the widely divergent response to antibiotic therapy, treatment needs to be individualized. This means that some patients may require much longer treatment with oral and/or intravenous antibiotics.

10. There is not sufficient evidence from the studies published to date to develop treatment guidelines.

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[12] Klempner et al. Two controlled trials of antibiotic treatment in patients with persistent symptoms and a history of Lyme disease. N Engl J Med 345:85-92, 2001

[13] Asch et al. Lyme disease: An infectious and postinfectious syndrome. J Rheum 21:454-60, 1994

[14] Valesov et al. Long-term results in patients with Lyme arthritis following treatment with ceftriaxone. Infec 24:98-102, 1996

[15] Steere et al. Treatment of Lyme arthritis. Arth Rheum 37:878-88, 1994

[16] Schmidli et al. Cultivation of *Borrelia burgdorferi* from joint fluid three months after treatment of facial palsy due to Lyme borreliosis. J Infect Dis 1988;158:905-6.

[17] Haupl, et al. Persistence of B. burgdorferi in Ligamentous Tissue from a Patient with Chronic Lyme Borreliosis. Arthritis Rheum 1993; 36:1621-6.

[18] Pfister et al. Randomized comparison of ceftriaxone and cefotaxime in Lyme neuroborreliosis. J Infect Dis. 1991; 163(2): 311-318.

[19] Hassler et al. Pulsed High Dose Cefotaxime Therapy in Refractory Lyme Borreliosis (Letter) Lancet; 338:193.

[20] MacDonald et al. Clinical implications of delayed growth of the Lyme borreliosis spirochete, *Borrelia burgdorferi*. Acta Tropica 1991;48:89-94.

In spite of the paucity of data, two groups of physicians that treat Lyme disease independently developed peer-reviewed 'evidence-based' treatment guidelines using the same literature (**Appendix A**) to formulate their treatment guidelines. The Infectious Disease Society of America (IDSA) advocates a maximum of 30 days of oral or intravenous antibiotics and assumes that the remaining symptoms reflect a self-perpetuating autoimmune response[21]. The International Lyme and Associated Diseases Society (ILADS), which is composed of physicians from a variety of specialties who primarily treat Lyme disease, assume that the persistent symptoms reflect on-going infection and gauge the duration of treatment by the patient's individual clinical response. These physicians believe that there is insufficient evidence at this point to adopt standardized treatment protocols[22]

While each viewpoint has a strong underlying hypothesis, the scientific evidence supporting either viewpoint is equivocal. Outcomes research is limited and conflicting. The NIAID has only funded three double-blind, placebo-controlled treatment outcome studies for long-term treatment of persistent Lyme disease. The findings of two studies (Klempner and Krupp-**Appendix A**) are contradictory, with one indicating that continued treatment is beneficial for treating fatigue and the other indicating that it is not. The third NIAID-funded study (Fallon-**Appendix A**) has recently been completed and preliminary results support continued antibiotic treatment for patients with persistent Lyme disease. The findings of nine non-controlled studies (**Appendix A**) support continued treatment. The existence of limited or conflicting controlled studies is not uncommon in the practice of medicine.

When a variety of viable treatment options exist, therapy is decided by weighing the individual's risks and benefits. Use of antibiotics can be associated with side effects, allergic reactions, development of drug resistance and cost. The benefits of antibiotics are the relief from a severe multisystemic bacterial infection that is difficult to eradicate with short-term antibiotic treatment. Withholding adequate antibiotic treatment for late-stage Lyme disease (when it is known that Borrelia persist in many treated patients-see **Appendix B**) is analogous to the Tuskegee experiment performed by the Public Health Service[23], which has been widely criticized for the failure to adequately treat African American men with late-stage syphilis, another spirochete disease.

Insurance companies have adopted guidelines reflecting short-term treatment approaches, which are governed by cost-containment considerations. However, the legal standard of care for treating a condition is determined by the consensus of physicians who actually treat patients, not by treatment guidelines[24]. One survey found that 57% of responding physicians treat persistent Lyme disease for three months or more[25]. Fallon notes that for over 3400 patients screened for the Columbia University study of persistent

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[21] Wormser, et al., Practice guidelines for the treatment of Lyme disease. The Infectious Diseases Society of America. Clin Infect Dis, 2000. 31 Suppl 1: p. 1–14.

[22] The International Lyme and Associated Diseases Society (ILADS), Evidence-based guidelines for the management of Lyme disease. Expert Rev Anti-infect Ther, 2004. 2(Suppl): p. S1–S13.

[23] Jones, JH. Bad Blood: The Tuskegee Syphilis Experiment, , expanded edition (New York: Free Press, 1993).

[24] Hurwitz, B., Clinical guidelines and the law. BMJ, 1995. 311: p. 1517–1518

Lyme disease, the mean duration of IV treatment was 2.3 months and the mean duration of oral antibiotic therapy was 7.5 months[26]. In another survey, "50% of the responders considered using antibiotics for a time greater than one year in a symptomatic seropositive Lyme disease patient. Almost that same number would extend therapy to 18 months if needed."[27]

When more than one standard of care exists, the critical question becomes who decides the appropriate course of treatment for the patient. Under the medical ethical principle of autonomy, the treatment decision belongs to the patient. Hence, the American Medical Association requires that the physician disclose and discuss with the patient not only the risks and benefits of the proposed treatment, but also the risks and benefits of available alternatives[28]. Treatment choices involve trade-offs between the risks and benefits of treatment options that only patients, who know the kinds of risks they are willing to run and the types of quality of life outcomes that matter to them, are uniquely suited to make.

Insurance companies have placed the full weight of their economic clout behind the less expensive short-term treatment protocols. More expensive longer-term treatment options are discredited as "experimental" or "not evidence-based." The point, of course, is that the science underlying both the short-term and the longer-term treatment options is equally uncertain. It is estimated that only 20% of medicine practiced today is rooted in double-blind studies[29] The bulk of medicine today is practiced in the grey zone. Evidence-based medicine requires only that medicine be practiced in accordance with the evidence that currently exists, not that treatment be withheld pending research. As for the cost considerations, healthcare costs generally are lower when the patient's preference is supported[30].

In an ideal world, decisions would be based on strong scientific evidence, consensus opinion, and the views of the treating physician. However, seldom are all three available. A recent symposium by the National Institute of Health Care Management Research and Educational Foundation found a general consensus *that care should not be denied because evidence is limited, conflicting, or even non-existent.* Rather, decisions should be based on the best information available. It has been noted that:

> Much, if not most, medical care, even that which is generally accepted in the medical community, would be denied under an evidence-based standard because

[25] Ziska, et al. Physician preferences in the diagnosis and treatment of Lyme disease in the United States. Infection, 1996. 24(2): p. 182–6.
[26] Fallon, B.A., Testimony at public hearings in re Lyme disease for the State of Connecticut Department of Public Health. 2004: p. 134–153.
[27] Katzel, J., Is there a consensus in treatment of Lyme Borreliosis?, in Lyme Disease 1991 Patient/Physician Perspectives from the U.S. & Canada, L. Mermin, Editor. 1992.
[28] American Medical Association, Code of Medical Ethics.
[29] Hitt, J., The year in ideas: a to z.; evidence-based medicine., in New York Times (December 9, 2001, Sunday).
[30] Wennberg, et al. Geography and the debate over Medicare reform. Health Aff (Millwood), 2002. Supp Web Exclusives: p. W96-114.

> so few health care services have been subject to rigorous research. At particular risk for denial of needed services are disabled persons because of the lack of treatments proven effective through clinical trials." (Independent Review of Managed Care Decisions by Honorable Mary C. Morgan. (Retired.)

Most patients who require prolonged intravenous antibiotics are denied coverage and subsequently undergo an independent medical review as part of the appeal process. First, it is imperative that those responsible for performing independent medical reviews be made aware of the fact that there are two recognized treatment approaches and that both sets of treatment guidelines be used as part of the review process. Second, the reviewers need to consider all of the data that illustrate the variability of treatment approaches physicians treating persistent Lyme Disease use (see **Appendix A** for the references). Third, the view of the treating physician needs to be given more weight, given that treatment outcomes research to date illustrate that the population being studied is enormously heterogeneous. In these situations, the clinical course of the individual patient is more a predictor of response to treatment than heterogeneous group studies. Fourth, the variation in treatment practices that currently exists should be resolved by promoting more outcomes research to help resolve the scientific uncertainty and patient's preference should be supported.

There are a number of ways that medical necessity may be determined: on facts and evidence, on a consensus of medical opinion, or on the judgment of individual physicians. Where outcomes research is limited or equivocal, decisions should be based on the best information available—which in the case of heterogeneous populations may well be the unique clinical course of the individual patient.

Appendix A
Summary of Clinical Studies for Treatment of Late Stage Lyme Disease

Randomized placebo-controlled Studies

Steere, 1985[31]-40 patients with established Lyme arthritis were randomized to receive weekly IM injections of benzathine penicillin or placebo. 35% of the treated patients had complete resolution of their symptoms and remained symptom-free during a mean follow-up period of 33 months. None of the placebo treated patients improved. As compared with nonresponders, penicillin responsive patients were more likely to have received antibiotics for early Lyme disease and less likely to have received intra-articular steroids.

Klempner, 2001[32]; Kaplan 2003[33]-78 seropositive (by Western Blot) and 51 seronegative patients with post-treatment Lyme disease were randomized to receive 30 days IV ceftriaxone followed by oral doxycycline 100mg bid or placebo for 60 days. Patients underwent standardized testing at baseline, 90 and 180 days. *There were normal baseline neuropsychological scores in all patients*. There were no significant differences between seropositive and seronegative patients in outcomes, nor were there significant differences between treated and untreated patients. Of note is the fact that 64% of patients had persistent symptoms after standard treatment for the disease. Thus, the validity of this study has been questioned.

Krupp, 2003[34]- Double blind placebo controlled trial on 55 patients with continued fatigue 6 or more months after antibiotic treatment (3 weeks oral Abs or IV ceftriaxone) for Lyme disease. Patients were randomized to receive placebo or 4 weeks of IV ceftriaxone. Outcome measures were fatigue, cognitive speed, and clearance of OspA antigen from the CSF. 64% of patients given antibiotics were improved compared with 18.5% given placebo. Further, for patients with positive western blots at baseline, the responder rate was 80% vs 13%. For seronegative patients, the responder rate was 46% vs 27%. Patients receiving antibiotics also had significantly lower pain scores than those receiving placebo. There were no differences between groups in results of neurocognitive tests. The authors concluded that repeated antibiotic therapy had a substantial positive effect on late Lyme disease outcome.

Fallon, 2004[35]-completed a trial of 10 weeks of IV antibiotic therapy in patients with late Lyme disease symptoms who had previously been treated with at least 3 weeks of IV antibiotics and then relapsed. There was significant improvement in cognition and

[31] Steere et al. Successful parenteral penicillin therapy of established Lyme arthritis. N Engl J Med 312:869-74, 1985
[32] Klempner et al, Two controlled trials of antibiotic treatment in patients with persistent symptoms and a history of Lyme disease. N Engl J Med 345:85-92, 2001
[33] Kaplan et al, Cognitive function in post-treatment Lyme disease: Do additional antibiotics help? Neurology 60:1916-1922, 2003
[34] Krupp et al. Study and treatment of post Lyme disease. A randomized double masked clinical trial. Neurology 60:
[35] In preparation.

other symptoms. This study was part of a $4.7 million NIH funded study. The manuscript is in preparation.

Randomized Trials

Dattwyler, 1988[36]-23 patients with clinically active late Lyme disease were randomly assigned to IV treatment with either penicillin or ceftriaxone. Of the 10 treated with penicillin, 5 were judged treatment failures; of the 13 who received ceftriaxone, only 1 patient did not respond. An additional 31 patients were subsequently treated with ceftriaxone with similar results. Patients that were unresponsive to ceftriaxone were more likely to have received corticosteroid treatment.

Pfister, 1989[37]-21 patients with radiculitis or neuroborreliosis associated with Lyme disease were randomized to receive a 10 day treatment with either IV penicillin G or cefataxime. There were no differences in the outcomes of the two groups. See Pfister, 1991 below.

Hassler, 1990[38]-135 patients with late-stage Lyme disease were randomized to receive IV penicillin G or IV cefotaxmine for 10 days. Cefotaxamine was significantly more effective than penicillin G with 87.9% vs 61.3% reporting full or partial remission of symptoms 24 months later.

Pfister, 1991[39]-33 patients with Lyme neuroborreliosis were randomized to receive a 10 day course of either IV ceftriaxone or Cefotaxime. Neurologic symptoms improved or subsided in 26/30 patients-there was no difference in treatment groups. At a mean follow-up of 8 months, 17/27 patients were clinically asymptomatic. Bb was isolated from the CSF of one patient 7 months after ceftriaxone therapy. Since 10 patients remained symptomatic, the authors concluded that a prolongation of therapy might be necessary.

Steere, 1994[40]-38 patients with Lyme arthritis were randomly assigned to 30 days of treatment with either doxycycline or amoxicillin plus probenecid. Patients who had persistent arthritis 3 months following treatment were given IV ceftriaxone for 2 weeks. 16/18 of the patients treated with amoxicillin and 18/20 treated with doxycycline had resolution of arthritis symptoms within 3 months of treatment. However, neuroborreliosis later developed in 5 patients. Of 16 patients with persistent arthritis who were treated with IV ceftriaxone, none had resolution of arthritis within 3 months. The authors concluded that even with resolution of specific manifestations of Lyme disease

[36] Dattwyler et al. Treatment of late Lyme borreliosis-randomized comparison of ceftriaxone and penicillin. Lancet, May 28: 1191-4, 1988
[37] Pfister et al. Cefotaxime vs penicillin F for acute neurologic manifestations in Lyme borreliosis. A prospective randomized study. Arch Neurol 46:1190-4, 1989
[38] Hassler et al. Cefotaxime versus penicillin in the late stage of Lyme disease-prospective, randomized therapeutic trial. Infection 18:16-20, 1990
[39] Pfister et al. Randomized comparison of ceftriaxone and cefotaxime in Lyme neuroborreliosis. J Infect Dis 163:311-18, 1991
[40] Steere et al. Treatment of Lyme arthritis. Arthritis Rheum 37:878-88, 1994

with oral antibiotics, there is still a risk of developing additional symptoms of Lyme at a later time. Persistent arthritis may be related to an autoimmune phenomenon (although they did not rule out persistent infection with PCR or culture). Others concluded that 2 weeks of IV ceftriaxone may be insufficient to address Lyme arthritis.

Wahlberg, 1994[41]-100 consecutive late-Lyme disease patients were treated with different antibiotic regimens and followed up for 12 months after treatment. Treatment outcome was successful in 4/13 patients treated with IV ceftriaxone for 14 days, 50/56 patients treated with ceftriaxone followed by 100 days of amoxicillin with probenecid, and 19/23 of those treated with IV ceftriaxone for 14 days followed by 100 days of cephadroxil.

Okski, 1998[42]-randomized 60 patients with disseminated Lyme borreliosis based on CDC diagnostic criteria to receive either cefexime and probenecid orally for 100 days or ceftriaxone IV for 14 days followed by oral amoxicillin and probenecid for 100 days. The immediate outcome after antibiotics was not different between the two treatment groups. However, after a year of follow-up, there were significantly greater relapses, treatment failures, and positive PCR tests. The results of this study support the use of intravenous antibiotics along with prolonged antibiotic therapy in patients with late-stage Lyme disease.

Fallon, 1999[43]-studied 23 Lyme patients who complained of persistent memory difficulties after IV antibiotic therapy of 4-16 weeks. Four months after their initial treatment, 18 of the patients received additional IV antibiotics and compared with the others who did not receive additional antibiotics. Those receiving additional IV antibiotics scored better on cognition tests, greatest functional improvement in energy, pain, and physical functioning than untreated patients. Based on the results from this pilot study, the authors concluded that there was enough evidence to plan a larger study investigating the utility of repeated courses of IV antibiotics (see Fallon, 2004).

Logigian, 1999[44]-a series of 18 consecutive patients with Lyme encephalopathy and symptoms of memory difficulty, minor depression, somnolence, or headache were treated with 30 days of IV ceftriaxone. At the beginning of treatment, 89% had abnormal memory scores, 89% had CSF abnormalities and all tested had perfusion defects on SPECT scan. Six months after treatment, memory scores were significantly improved, CSF protein levels were significantly less, and post-treatment perfusion was significantly improved. 12-24 months after treatment, all patients rated themselves as improved or back to normal.

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[41] Wahlberg et al. Treatment of late Lyme borreliosis. J Infec 29:255-61, 1994

[42] Oski, et al. Comparison of oral cefixime and intravenous ceftriaxone followed by oral amoxicillin in disseminated Lyme borreliosis. Eur J Clin Microbiol Infec Dis 17:715-19, 1998.

[43] Fallon et al. Repeated antibiotic treatment in chronic Lyme disease. J Spirochetal Tick Dis 6:94-101, 1999

[44] Logigian et al. Successful treatment of Lyme encephalopathy with intravenous ceftriaxone. J Infec Dis 180:377-83, 1999

Observational studies

Hassler, 1991[45]-reported on two patients with antibiotic resistant Lyme disease that were treated with pulsed high-dose cefataxime with 2 days of treatment followed by 6 days without antibiotics over a ten-week period of time. One patient was symptom-free 6 months after antibiotic treatment, the other was improved and skin biopsies showed no evidence of Borrelia.

Cimmino, 1992[46]-reported 2 cases of chronic Lyme arthritis, refractory to standard antibiotic treatment, who were treated with

Valesov, 1996[47]-reported on the outcome of a 36-month follow-up of patients with late stage Lyme arthritis after 2 weeks of ceftriaxone therapy. At 36 months 19/26 continued to be symptom-free, 6 had relapsed and 1 presented with new late-Lyme symptoms.

Donta, 1997[48]-277 patients with chronic Lyme disease and symptoms of fatigue, musculoskeletal pain, neuropsychiatric dysfunction, and paresthesias were treated with tetracycline for 1-11 months (mean 4 months). Overall 20% of the patients were cured, 70% significantly improved (degree of improvement 75-100%), and 10% did not improve. Improvement frequently did not take place for several weeks: after 2 months of treatment, 33% were significantly improved, after 3 months 61% were significantly improved. Improvement showed as early as one to two weeks after the start of treatment; however, in patients who were symptomatic for more than a year, it frequently took 4-6 weeks on the antibiotic for evidence of improvement. This slow rate of improvement was postulated to be due to the slow rates of multiplication and metabolism in Borrelia. This study underscores the necessity of an individualized approach to the treatment of late-stage Lyme disease.

Okski, 1999[49]-13 patients who had clinical relapses and were PCR positive after at least 3 months of oral antibiotics were treated with IV ceftriaxone for 4-6 weeks. None of the patients were PCR positive after treatment and 9 showed good therapeutic responses. The authors concluded that treating late Lyme disease with appropriate antibiotics for more than 3 months may not always eliminate Borrelia and that longer courses may be necessary.

Donta, 2003[50]-235 patients with chronic Lyme disease symptoms of fatigue, musculoskeletal pain, and neurocognitive dysfunction with positive serology for Borrelia

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[45] Hassler et al. Pulsed high-dose cefotaxime therapy in refractory Lyme borreliosis. The Lancet 338:1991
[46] Cimmino et al. Long-term treatment of chronic Lyme arthritis with Benzathine penicillin. Ann Rheum Dis 51:1007-8, 1992
[47] Valesov et al. Long-term results in patients with Lyme arthritis following treatment with ceftriaxone. Infection 24:98-102, 1996
[48] Donta ST. Tetracycline therapy for chronic Lyme disease. Clin Infect Dis 1997; Suppl 1:S52-S56
[49] Oksi et al. Borrelia burgdorferi detected by culture and PCR in clinical relapse of disseminated Lyme borreliosis. Annals of Medicine 31: 225-32, 1999b
[50] Donta ST. Macrolide therapy of chronic Lyme disease. Med Sci Monit 9:136-142, 2003

were treated with macrolide antibiotics and hydroxychloroquine for one or more months based on their level of improvement during the course of treatment. 120 patients who were improved at the discontinuation of therapy had relapsing symptoms and were retreated with antibiotics. Of those retreated with macrolide/hydroxychloroquine, 32/33 had improvement, tetracycline 54/74 had improvement, IV ceftriaxone 9/23 had improvement. Thus, tetracycline and IV ceftriaxone had a much lower success rate than macrolide/hydroxychloroquine therapy.

Reviews

Cimmino, 1996[51]-reviewed the results of antibiotic treatment of Lyme arthritis in peer-reviewed journals between 1985 and 1991. The studies were small or medium-sized

[51] Cimmino et al. Treatment of Lyme arthritis. Infection 1:91-94, 1996

I have read or will read this week all the articles above. I understand my care does not fit the position of the Infection Society of America or most Infection Specialists.

SIGN_____

DATE_____

PRINT

NAME_____

FAMILY SUPPORT INFORMED CONSENT

In Dr. Schaller's practice it is important that your spouse, siblings and parents, if any are applicable, support your care and the treatment's Dr. Schaller is offering. If they oppose your treatment this problem has to be resolved in 10 days. Why?

First, you will not be able to have a solid physician-patient relationship; if others are whispering in your ear "Dr. Schaller is wrong." This type of criticism causes poor compliance with treatment and can lead to a bad end to the patient-MD relationship.

Since ***ill patients always have a small risk of a clot*** causing death or other risks found in those who are ill, it increases an unacceptable malpractice burden if a bad medical incident happens when relatives are unsupportive of Dr. Schaller's treatment. Alienated relatives suffering understandable loss look to someone as "the cause" of the injury or death. While this type of problem has yet to happen in Dr. Schaller's practice, it eventually will happen.

Dr. Schaller tends to treat large families and extended families and clans. One person comes and gets better and then other's come. This is great fun and satisfying. If the clan is opposed to him, this is a problem in a state like Florida that ends a doctor's medical career if a doctor has three malpractice cases.

Since Dr. Schaller is often NOT using basic and common labs, nor using routine and standard treatments, it will be easy for relatives to accuse him of "acting outside of the standard of medicine." Dr. Schaller feels in some areas that is a compliment, and is true, but the courts sometimes do not support innovative treatment. Angry relatives can be litigious relatives.

Dr. Schaller has many books and research papers in many areas of medicine and does not feel that he should have to "prove" or "sell" himself to anyone **who is a thinking person.** If a relative looks at his web sites and publications and is still suspicious, defamatory or unsupportive, Dr. Schaller is not able to not have any involvement in your care.

Some specific areas of concern in the past are listed and are not acceptable:

1) A husband reads a couple lay level basic articles on your illness and is unsupportive of your treatment. This has to be resolved and would require a meeting to discuss my approach. If the spouse is still not supportive, I cannot treat you.

2) Parent's say they want to help an **adult** child and then after a few sessions they are not satisfied since you are not fully "cured" and stop the financial support for the treatment. This type of financial manipulation and control is unacceptable, and would mean your care would be transferred.

3) The "know it all" relative who has a friend in the healthcare field or is in the healthcare field. They hear the results of your intake consult and defame the diagnosis and plan. Since many of Dr. Schaller's patient's have been to 3-30 physicians before him, the basic solutions have already failed. **The know-it-all relative will need to be renounced by the patient in a written note**, "I do not support any of the comments made by my Uncle Tom, who is a retired surgeon, because he does not understand the issues involved. I have heard his 'case' and opinion's and reject his counsel and oppose his criticism's of Dr. Schaller.'

4) Amazingly, some character disordered family members have actually threatened Dr. Schaller in various ways. This will result in an immediate end of the physician-patient relationship. Though you will be given medications, if applicable, for a full month and offered referrals if you want them. Threats of any kind will end any physician-patient relationship.

5) It is fine to say Dr. Schaller "may have made a mistake," if you are fairly certain he did, and **we will address and fix this as fast as possible.** Though, please understand Dr. Schaller takes careful notes and sometimes people's memories are limited. The most common example is in the prescribing of a medication. Pharmacies lose prescriptions that are called in about 1/40x. Do not assume Dr. Schaller did not call it in a prescription, if he clearly said he would. On occasion he forgets, at a rate of about 1/700, and he will certainly admit it if his notes show he forgot to call in a prescription. He will fix his error.

   Also, some sincere patients assume Dr. Schaller would do something he did not agree to do. The most common example is that a medication is **mentioned briefly** and the patient assumes it was called in**. But Dr. Schaller does not call in any medication until you have been given 1-3 handouts on the medication or you have found 1-3 articles on the medication from the Internet**, so you are **educated** and can use the medication in **an "informed" manner.** This is good medical care and makes you smart about the side effects and interactions of any treatment.

In summary, we have a very friendly and cozy practice. People are generally very pleased with our wide and broad range of services and care. We do not want to be involved in knowledge debates and hostilities, and if these

emerge we need to address them promptly.

Why such concern over this issue?

Angry, bitter or know-it-all relatives and friends ruin the pleasure of the physician-MD team. They increase malpractice risks. It makes working in medicine unpleasant and joyless. It also shows that the family is clue-less about the services and scholarship of Dr. Schaller, and this means they are more likely to see him as **a cause** of a bad event, then as a solid clinician researcher who is trying to offer top medical care.

I agree to inform Dr. Schaller *promptly and within 10 days* if any of my relatives or friends oppose his diagnoses or treatments. I understand I may be asked to allow Dr. Schaller to talk to the person disrupting care. If Dr. Schaller and my relative have polar positions on treatment, I will be asked to choose in writing that I support Dr. Schaller's care or I will be offered the option of having one or more other physician's replace Dr. Schaller.

I HAVE READ THE ABOVE.

NAME SIGNED_____DATE_____
NAME PRINTED_____

ALIENATION AGREEMENT
WITH DR. SCHALLER

In the past some people with limited means have hired Dr. Schaller to treat them. Some have asked for a simple diagnostic consult over 2 sessions with labs and then have gone on their way, satisfied they have new ideas to fix their troubles. Others have tried to remain patients who could not afford monthly sessions for active illness, and then tried to

rush sessions and on rare occasssions some have come to resent Dr. Schaller for his fees. Dr. Schaller is a full-time researcher. He does not get rich spending most of his time reading hundreds articles a week. He only sees patients 19 hours a week or less. He would make much more with lower rates and seeing 40 hours of patients a week, but he is looking for new solutions and looking at the work of hundreds of researchers to learn weekly and aggressively.

His dream is that one of his inventions or actions might allow him to treat for free or at profoundly low fees. That is not remotely possible now with his self-funded intense study practice.

If you begin to feel ANY RESENTMENT WITH DR. SCHALLER FOR ANY REASON OR HIS STAFF YOU ARE REQUIRED TO MENTION IT IN A WEEK. WE DO NOT WANT TO HEAR ABOUT THE LAST STRAW. WE WANT TO KNOW ABOUT THE FIRST STRAY.

If you cannot afford ongoing services, watch you resentment and expectations. If you leave on good terms, you can do NON-PATIENT consults any time you want.

DO YOU AGREE TO THE ABOVE?

SIGN_____ PRINT_____DATE_____

Infectious Disease Antimicrobial Therapy:

Overview of Antimicrobial Therapy

Timothy H. Dellit, MD; Thomas M. Hooton, MD, FACP

[Infections Have Risks and Antibiotic Treatments Have Risks]

The essential feature of effective antimicrobial agents is the ability to inhibit the growth of microorganisms at concentrations tolerated by the host. Antimicrobial agents generally target anatomic structures or biosynthetic pathways unique to microorganisms.

The appropriate choice of an antimicrobial for an infection depends on five considerations: (1) the infecting organism and its antimicrobial susceptibilities; (2) the type of infection (e.g., abscess, bacteremia, meningitis, or urinary tract infection); (3) host factors (e.g., neutropenia, immune deficiencies, concurrent illnesses, age, drug allergies, and renal function); (4) factors associated with antimicrobial agents (e.g., dosage, routes of administration, drug interactions, serum levels and tissue penetration, potential toxicities, and cost); and (5) public health considerations. The widespread use of antibiotics has led to selection for highly resistant organisms that subsequently pose a risk for the patient and the community.

Identification of the Infecting Organism

Prompt identification of the causative organism is essential for the selection of appropriate antimicrobial drugs. A Gram stain of infected body fluids can provide early clues to the etiologic agent. Other relatively simple laboratory procedures, such as immunodiagnostic tests, may provide rapid identification of infecting organisms. If it is not possible to perform a Gram

stain, initial decisions are based on the clinical features and the usual bacteriology of the illness, pending results of appropriate cultures. Culture results must be interpreted with full recognition of the indigenous flora on various mucosal surfaces and on the skin. Culture identification of the infecting organism and knowledge of the local antibiograms can offer clues to the likely antimicrobial susceptibilities [*see Table 1 -- omitted*]. However, even within a single institution's antibiogram, patterns of susceptibilities are influenced by patient characteristics (e.g., previous antimicrobial exposure) and location within the hospital (e.g., intensive care unit or general ward). National trends in antimicrobial resistance are tracked by the Centers for Disease Control and Prevention (CDC)
(http://www.cdc.gov/drugresistance/community).

Determination of Bacterial Susceptibility to Specific Drugs

Determination of the in vitro susceptibility of isolated bacteria is usually performed to ensure that the empirical antimicrobial regimen is optimal. Timely availability of susceptibility data allows tailoring, when appropriate, of the antimicrobial regimen. However, the susceptibility of certain species (e.g., group A streptococci) to the drug of choice (e.g., penicillin G) has been so uniform that routine in vitro testing is unnecessary. Moreover, in vitro susceptibility data may not accurately predict the therapeutic outcome. For example, *Enterobacter* infections treated with third-generation cephalosporins, such as ceftazidime, may lead to induction of AmpC β-lactamases and clinical failure despite initial in vitro susceptibility.[1] Similarly, *Klebsiella* species and *Escherichia coli* may harbor extended-spectrum β-lactamases (ESBLs) and still appear susceptible to cephalosporins in vitro unless the clinical microbiology laboratory actively screens for their presence. Cephalosporin treatment of such an apparently susceptible organism with an unrecognized ESBL, however, is associated

with a high rate of failure.[2]

Disk diffusion is the classic method of testing antimicrobial susceptibility, although more automated systems are now available. When necessary, organisms can be tested by tube dilution techniques to determine the minimal inhibitory concentration (MIC) and the minimal bactericidal concentration (MBC) of various antibiotics. Newer techniques include the gradient diffusion method and various nucleic acid-based tests that can detect genes conferring antimicrobial resistance in bacteria.[3]

The Site of Infection and Ancillary Therapy

The efficacy of antimicrobial therapy depends on drug delivery to the site of infection. Transport across the blood-brain barrier varies considerably among antibiotics.[4] Intracellular parasites, such as rickettsiae and mycobacteria, require therapy with agents that can reach the intracellular milieu of the organism. Necrotic lesions, such as sequestra in bone or large abscesses, hinder penetration by antimicrobials. Drainage, debridement, or both, combined with antimicrobials, are necessary. Antimicrobial therapy for infection in or around a foreign body or prosthesis often is ineffective unless these materials are removed or replaced. Infections behind obstructing lesions, such as pneumonia that occurs behind a blocked bronchus or cholecystitis caused by biliary obstruction, will not respond to antibiotics until the obstruction is relieved.

Factors Affecting Dosage and Route of Administration

Many antimicrobials are absorbed sufficiently well via the oral route to provide effective serum levels in patients with normal gastrointestinal function.[5] For some antimicrobials (e.g., fluoroquinolones, metronidazole, and linezolid), the pharmacokinetics of the oral agents are almost the same

as those of the parenteral formulations, and oral therapy, when appropriate, is much less expensive. The enteral absorption of some antimicrobial agents is impeded by food and some medications (e.g., antacids); these antibiotics, such as tetracyclines, should be administered at least 1 hour before meals or several hours after meals. Certain antibiotics, such as neomycin, are essentially nonabsorbable and are used primarily for their effects on bowel flora. Intramuscular administration is used in a small number of conditions (e.g., benzathine penicillin for syphilis or ceftriaxone for gonorrhea), but most antimicrobials cannot be given intramuscularly because of local pain or necrosis at the injection site. Intravenous administration must be used in the treatment of major and life-threatening infections such as septic shock, meningitis, and endocarditis. Patients who are clinically stable after initial intravenous treatment may be ready for discharge from the hospital and may be able to receive parenteral antibiotics on an outpatient basis. Newer antimicrobial agents with long half-lives can be administered with improved intravenous catheters and delivery devices in simplified regimens. Alternatively, the enhanced bioavailability of some antimicrobials has allowed many patients to complete a course of oral antimicrobial therapy after they are discharged. Such pharmacologic advances have led to substantial economic benefits, enhanced patient comfort, and good therapeutic results with few complications.[6]

Host Factors

Concurrent Illnesses

Patients with immunosuppressive illnesses are vulnerable to opportunistic pathogens. These patients may require broader antimicrobial coverage and combination therapy for ordinary pathogens. The same is true to a lesser extent for patients with chronic debilitating illness. Patients with renal

insufficiency or liver disease may be unusually susceptible to direct drug toxicity [*seeDirect Drug Toxicity -- omitted, below*].

Many antimicrobials (e.g., penicillins, cephalosporins, aminoglycosides, vancomycin, and fluoroquinolones) are excreted primarily by the kidneys. Guidelines for dosage adjustment in patients who have severe renal failure and are undergoing dialysis or peritoneal dialysis are provided elsewhere [*see 10:IX Pharmacologic Approach to Renal Insufficiency, Appendix A -- omitted*]. Antibiotics with nonrenal metabolism may be preferable in azotemic patients, provided that the infecting organism is susceptible. A number of antibiotics, including nafcillin, piperacillin, doxycycline, metronidazole, and rifampin, are excreted by the liver.

Pregnancy

The administration of antimicrobials during pregnancy and in the postpartum period poses several problems. Foremost is the question of safety, both for the mother and for the fetus or neonate. Although most antibiotics cross the placenta and enter maternal milk, the concentrations to which the fetus or neonate is exposed vary widely. Because the immature liver may lack the enzymes required to metabolize certain drugs, pharmacokinetics and toxicities in the fetus are often very different from those in older children and adults. Teratogenicity is a major concern when any drug is administered during pregnancy. Finally, it may be necessary to alter the dosage schedules of drugs that appear to be safe to use during pregnancy. Increases in maternal blood volume, glomerular filtration rate (GFR), and hepatic metabolic activity often reduce the maternal serum levels of antimicrobials by 10% to 50%, especially late in pregnancy and in the early postpartum period. In some women, delayed gastric emptying may reduce the absorption of antibiotics that have been administered orally

during pregnancy.

Even though 25% to 40% of women receive antibiotics during pregnancy, data regarding safety and efficacy in this setting are often scarce. Some general recommendations have been proposed, but they are intended only as a guide [*see Table 2 -- omitted*]; in all cases, therapy must be individualized, and both the indications for antibiotics and the possible risks to mother and fetus must be considered. Individual decisions are also required for lactating mothers. Although most antimicrobials appear safe for breast-fed infants, chloramphenicol, fluoroquinolones, and tetracyclines should be avoided.[7]

Advanced Age

Physiologic changes that occur with age can alter the pharmacokinetics of antimicrobials. For example, decreased gastric acidity and intestinal motility can impair drug absorption; increased body fat and decreased serum albumin levels can alter drug distribution; and decreased hepatic blood flow and enzymatic action can delay drug metabolism. Although these factors have not consistently affected antibiotic levels in the elderly, the decrease in GFR that occurs with age can lead to the accumulation of drugs excreted by the kidney. The high therapeutic index of the penicillins and cephalosporins obviates major changes in dosage schedules in elderly patients who have normal serum creatinine levels. However, in the case of aminoglycosides and vancomycin, decreased dosage schedules are often required. Ideally, drug levels should be measured and renal function should be monitored when these agents are given.

Complications of Antimicrobial Therapy

There are three general types of complications with antimicrobials: hypersensitivity reactions (which are not dose related), direct drug toxicity

(which usually is dose related and manifests itself in a single organ or, occasionally, in several organs), and microbial superinfection (which is caused by drug-resistant organisms, such as *Clostridium difficile*, that flourish under the selective pressures exerted by antimicrobial therapy). In rare situations, antimicrobial therapy can lead to a worse clinical outcome. For example, antibiotic therapy has been associated with the development of hemolytic-uremic syndrome in children with gastroenteritis caused by *E. coli* 0157:H7.[8]

Hypersensitivity Reactions

A history of allergies should be taken before antimicrobial therapy is initiated in any patient.[9] More information is available regarding allergies to the penicillins than allergies to other agents, but skin eruptions, drug fever, and even anaphylaxis may be produced by many antibiotics. Allergic reactions occur in 1% to 10% of patients who receive penicillin. Fatal anaphylactic reactions are much less frequent [*seeAllergic Reactions to β-Lactam Antibiotics -- omitted, below*].

Direct Drug Toxicity

Although antimicrobials can damage virtually all human organ systems, the potential for toxicity varies widely from drug to drug.[10] The principal antibiotics that are directly toxic to the kidney are aminoglycosides and polymyxins. Patients with preexisting renal insufficiency are at increased risk for toxic reactions to various antibiotics, including nephrotoxicity, coagulopathies and other hematologic toxicities, seizures, and ototoxicity and other neurotoxicities.

Hematologic [Blood] toxicity. Chloramphenicol can produce bone marrow suppression, which is usually mild and reversible, but rarely presents as an

irreversible fatal aplastic anemia. Chloramphenicol, sulfonamides, nitrofurantoin, and primaquine can cause hemolytic anemia in patients who have deficiencies of erythrocyte glucose-6-phosphate dehydrogenase (G6PD). Hemolytic anemia, thrombocytopenia, and leukopenia that involve an immune mechanism can be caused by penicillins, cephalosporins, tetracyclines, and rifampin, but these reactions are uncommon. Macrolides and trimethoprim-sulfamethoxazole have been associated with agranulocytosis. Trimethoprim can produce anemia, leukopenia, and thrombocytopenia from folate deficiency, which is reversible with folinic acid. Linezolid can also produce myelosuppression, which appears to be reversible with discontinuance of the drug. Patients receiving linezolid for more than 2 weeks are at greatest risk, and the complete blood count with differential should be monitored weekly in such patients. Neutropenia can occur during therapy with penicillins, cephalosporins, or vancomycin. It may be severe but is self-limited, with recovery occurring 1 to 7 days after the antibiotic is withdrawn. Penicillins inhibit platelet aggregation by adenosine diphosphate, which may account for the bleeding that occurs in some patients receiving these antibiotics in high doses. Various cephalosporins may produce coagulopathies by prolonging the prothrombin time. The methylthiotetrazole side chain present in cephalosporins such as cefotetan appears to be responsible.

Ototoxicity. [Ear Damage] Antibiotics may produce a wide range of toxic effects on the central and peripheral nervous systems. Ototoxicity, either vestibular or auditory, can be produced by any of the aminoglycosides; neuromuscular blockade is much less common. Minocycline has occasionally been reported to produce significant vestibular reactions. Vancomycin can cause auditory neurotoxicity. Intravenous administration of penicillin and other β-lactams may produce seizures, especially when administered in very high doses or when given to azotemic patients or to

patients with underlying epilepsy.

Neurotoxicity. Peripheral neuropathy can occur as a complication of therapy with nitrofurantoin, particularly when renal failure is present. The neuropathy that occurs with isoniazid can be prevented by the daily administration of 50 mg of pyridoxine. Tetracycline may in rare instances produce reversible benign intracranial hypertension with headache and papilledema. Nalidixic acid may also produce intracranial hypertension and seizures in children. Metronidazole can sometimes cause ataxia, encephalopathy, seizures, or peripheral neuropathies. Ofloxacin has been reported to cause seizures, and mania has been attributed to clarithromycin. Optic neuritis, usually manifested by decreased visual acuity and decreased perception of the color green, may occur as a side effect of ethambutol.

Hepatotoxicity. [Liver] The principal antimicrobials that produce adverse effects on the liver are those used in the treatment of tuberculosis: isoniazid, rifampin, aminosalicylic acid, and pyrazinamide. Trovafloxacin was withdrawn because of hepatotoxicity, but other fluoroquinolones have not been implicated. The tetracyclines can occasionally cause fatty liver with hepatotoxicity; this is most likely to occur in patients receiving 2 g or more daily by the intravenous route. Patients receiving high-dose β-lactam antibiotics may develop hepatitis or cholestasis, presumably as a result of hypersensitivity reactions. Nitrofurantoin may cause chronic active hepatitis in some patients. Erythromycins and sulfonamides have been associated with acute hepatitis.

GI complications. GI reactions to antibiotics result either from direct irritation by the drug, the occurrence of which is usually dose related, or from bacterial overgrowth.[11] Irritative GI side effects are usually produced when antibiotics are administered orally rather than parenterally. The

predominant site of irritation varies from drug to drug; for example, erythromycin more commonly produces gastric irritation with epigastric distress and nausea, whereas tetracyclines may produce diarrhea and upper GI symptoms. Some qualitative and quantitative changes in the intestinal flora occur after antibiotic administration. These intestinal changes may contribute to flatulence and other lower GI symptoms, including antibiotic-associated diarrhea, which are quite common when broad-spectrum antibiotics are administered orally. An important subset of antibiotic-associated diarrhea involves the selective overgrowth of *Clostridium difficile* and resultant pseudomembranous colitis [*see 7:V Anaerobic Infections -- omitted* and *4:III Diarrheal Diseases -- omitted*].

Other toxicities. Antibiotics may cause various other toxicities. Erythromycin and other macrolides can cause prolongation of the QTc interval and polymorphic ventricular tachycardia. In rare instances, this toxicity occurs in the absence of predisposing factors, but it is more likely to occur in patients with significant heart disease and in patients taking terfenadine, astemizole, or cisapride. Several fluoroquinolones, particularly grepafloxacin and sparfloxacin can have similar effects on cardiac conduction.[12] All fluoroquinolones can cause tendonitis and either hyperglycemia or hypoglycemia. Trimethoprim-sulfamethoxazole can cause hyperkalemia [low potassium which can harm heart], particularly in azotemic patients. Sulfonamides, fluoroquinolones, and tetracyclines can produce photosensitivity.[13]

Microbial Superinfection

Antimicrobial therapy reduces the number of susceptible organisms from the normal flora of the skin, oropharynx, genitourinary tract, and gastrointestinal tract and exerts selective pressures that favor survival of drug-resistant

organisms. Such resistant organisms can occasionally establish a superinfection (i.e., a new infection caused by a different pathogen from the one being treated) either at the site of the original infection or at remote sites. Recent reports of both the increasing incidence and severity of *C. difficile* infections indicate that further caution is warranted against the indiscriminant use of antimicrobials, particularly broad-spectrum cephalosporins and fluoroquinolones.[14]

Public Health Considerations

Antimicrobial Resistance

The extensive use of antimicrobials, especially in intensive care units[15] and other health care facilities, strongly favors the selection of resistant microbial species, particularly bacterial strains harboring plasmids that confer transmissible resistance.[16–18] The widespread use of antibiotics in animal husbandry and agriculture compounds the problem; roughly 50% of the 25,000 tons of antibiotics that are sold annually in the United States are used in agriculture and aquaculture.[19] Infections from highly resistant strains of *Enterococcus, Streptococcus pneumoniae, Staphylococcus, Neisseria gonorrhoeae, Salmonella, Serratia, Klebsiella, Acinetobacter*, *Enterobacter, Pseudomonas*, and *Mycobacterium* have become important problems. Infections from resistant strains can spread rapidly—first within an institution, then throughout a community, and eventually even globally.[20] Although antimicrobial resistance is a worldwide problem, control depends on local measures, beginning with the judicious prescription of antibiotics by individual practitioners in combination with sound infection control practices. Through the Campaign to Prevent Antimicrobial Resistance in Healthcare Settings (http://www.cdc.gov/drugresistance/healthcare), the CDC offers resources for infection control. There is national concern that the

relatively small number of new antimicrobial agents in development will not keep pace with the emergence of drug resistance.

SIGN_____
DATE_____

Print Name_____

National Guideline Clearinghouse: Lyme Disease Guideline Comparison

- **GUIDELINE TITLE**Evidence-based guidelines for the management of Lyme disease.Practice guidelines for the treatment of Lyme disease.**DATE RELEASED**20042000 Jul**ADAPTATION**Not applicable: The guideline was not adapted from another source.Not applicable: The guideline was not adapted from another source.**GUIDELINE DEVELOPER(S)**International Lyme and Associated Diseases Society - Disease Specific SocietyInfectious Diseases Society of America - Medical Specialty Society**SOURCE(S) OF FUNDINGInternational Lyme and Associated Diseases Society (ILADS)Infectious Diseases Society of America (IDSA)COMPOSITION OF GROUP THAT AUTHORED THE**

GUIDELINEDaniel Cameron, MD, MPH, Internal Medicine and Epidemiology; Andrea Gaito, MD, Rheumatology; Nick Harris, PhD, Immunology; Gregory Bach, DO, Family and Integrative Medicine; Sabra Bellovin, MD, Family Practice; Kenneth Bock, MD, Family Practice; Steven Bock, MD, Family Practice; Joseph Burrascano, MD, Internal Medicine; Constance Dickey, RN, Registered Nurse; Richard Horowitz, MD, Internal Medicine; Steven Phillips, MD, Internal Medicine; Laurence Meer-Scherrer, MD, Internal Medicine; Bernard Raxlen, MD; Psychiatry; Virginia Sherr, MD, Psychiatry; Harold Smith, MD, Emergency Medicine; Pat Smith, President, Lyme Disease Association, Inc.; Raphael Stricker, MD, Hematology and Immunotherapy.*Authors*: Gary P. Wormser, Robert B. Nadelman, Raymond J. Dattwyler, David T. Dennis, Eugene D. Shapiro, Allen C. Steere, Thomas J. Rush, Daniel W. Rahn, Patricia K. Coyle, David H. Persing, Durland Fish, and Benjamin J. Luft.**CONFLICTS OF INTEREST**Not statedNot stated**DISEASE/CONDITION(S)**Lyme diseaseLyme disease**GUIDELINE CATEGORY**Diagnosis Evaluation Management TreatmentManagement Prevention Treatment**CLINICAL SPECIALTY**Emergency Medicine Family Practice Infectious Diseases Internal Medicine Nursing Psychiatry RheumatologyFamily Practice Infectious Diseases Internal Medicine Neurology Rheumatology**INTENDED USERS**Advanced Practice Nurses Nurses Physician Assistants Physicians Public Health DepartmentsAllied Health Personnel Physicians**GUIDELINE OBJECTIVE(S)**To serve as a resource for physicians, public health officials, and organizations involved in the evaluation and treatment of Lyme disease

- To present practitioners with practical and defensible guidelines for treating all individuals with Lyme disease including those with persistent, recurrent and relapsing symptoms of *Borrelia burgdorferi* infection.To provide clinicians and other health care practitioners with recommendations for the management of patients diagnosed with Lyme disease, or patients bitten by an *Ixodes* tick in North America**TARGET POPULATION**Patients presenting with symptoms associated with Lyme disease

- Patients diagnosed with Lyme diseasePatients with Lyme disease or patients bitten by an *Ixodes* tick in North America**MAJOR OUTCOMES CONSIDERED**In developing these treatment guidelines, the guideline developers considered factors such as incidence of Lyme disease; severity of disease in terms of morbidity; comorbidities and determinants of when Lyme disease is most likely to become chronic; feasibility, efficacy, and cost of antibiotic treatment; impact of antibiotic therapy on quality of life, including adverse drug events; and the potential for drug resistance to develop.Prevention of Lyme disease
- Prevention of other *Ixodes*-borne illnesses, including babesiosis and human granulocytic ehrlichiosis
- Resolution of symptoms and signs of early Lyme disease and prevention of late complications
- Effective treatment of late complications of Lyme disease while minimizing the adverse effects of antibiotic therapy
- Risks and consequences of developing Lyme disease

- Cost and adverse effects of antimicrobial therapy

Quality of life**COST ANALYSIS PERFORMED/REVIEWED? (YES/NO)**NoYes**METHODS USED TO COLLECT/SELECT EVIDENCE**Hand-searches of Published Literature (Primary Sources)
Hand-searches of Published Literature (Secondary Sources)
Searches of Electronic DatabasesSearches of Electronic Databases**DESCRIPTION OF METHODS USED TO COLLECT/SELECT THE EVIDENCE**English-language articles published from 1975 to 2003 were selected. The selection panel synthesized the recommendations from published and expert opinion. Human studies of Lyme disease were identified from MEDLINE (1975 to 2003) and from references in pertinent articles and reviews. Also included were abstracts and material presented at professional meetings and the collective experiences of the International Lyme and Associated Diseases Society (ILADS) Working Group treating tens of thousands of Lyme disease patients.Not stated**METHODS USED TO ASSESS THE QUALITY AND STRENGTH OF THE EVIDENCE**Weighting According to a Rating Scheme (Scheme Given)Weighting According to a Rating Scheme (Scheme Given)**RATING SCHEME FOR THE STRENGTH OF THE EVIDENCEQuality of the Data**

I. At least one randomized controlled trial supports the recommendation
II. Evidence from at least one well-designed clinical trial without randomization supports the recommendation

I. "Expert opinion"Evidence from at least one properly randomized, controlled trial
II. Evidence from at least one well-designed clinical trial without randomization, from cohort or case-controlled analytic studies (preferably from more than one center), from multiple time-series studies, or from dramatic results of uncontrolled experiments

Evidence from opinions of respected authorities based on clinical experience, descriptive studies, or reports of expert committees**METHODS USED TO ANALYZE THE EVIDENCE**ReviewReview of Published Meta-Analyses
Systematic Review**METHODS USED TO FORMULATE THE RECOMMENDATIONS**Expert ConsensusNot stated**DESCRIPTION OF METHODS USED TO FORMULATE THE RECOMMENDATIONS**Not statedNot stated**RATING SCHEME FOR THE STRENGTH OF THE RECOMMENDATIONS**Recommendations rated

A. **G**ood evidence to support the recommendation.
B. Moderate evidence to support the recommendation.
C. Optional.
D. Generally should not be offered.

Contraindicated.**Strength of recommendation**:

A. Good evidence to support a recommendation for use
B. Moderate evidence to support a recommendation for use

C. Poor evidence to support a recommendation
D. Moderate evidence to support a recommendation against use

Good evidence to support a recommendation against use**METHOD OF GUIDELINE VALIDATION**Peer ReviewPeer Review**DESCRIPTION OF METHOD OF GUIDELINE VALIDATION**Not statedNot stated**CLINICAL ALGORITHM? (YES/NO)** NoNo**IMPLEMENTATION PLAN DEVELOPED? (YES/NO)**NoNo**HAS PATIENT INFO? (YES/NO)**NoNo**VIEW MAJOR RECOMMENDATIONS**View Major Recommendations**View Major Recommendations**VIEW AVAILABILITY OF FULL TEXT**View Availability Information**View Full-text Guideline**

IV Treatment Consent Form [IF APPLICABLE]

In certain situations, IV medications are preferred or may be used to supplement oral medications. IV treatment is sometimes recommended when there is neurological involvement, carditis, complicated Lyme arthritis, or inadequate response to oral medications. Because the GI system is bypassed, IV dosing guarantees adequate medication blood levels and reduces gastrointestinal symptoms. IV antibiotics also may penetrate the blood brain barrier more effectively.

There are additional risks associated with IV treatment. IV treatment usually involves the installation of a PICC line in your vein to allow easier IV treatment. PICC lines may result in local infection

(less than 1% risk), or if unchecked, systemic infection. If signs of skin inflammation occur, the line may need to be removed. PICC lines rarely (less than 1% risk) may break off when they are being removed and may then require removal by a radiologist. One study suggests that the risks associated with IV antibiotics are estimated at 1 in 1,000 for major side effects and 3 in 100 for minor side effects.[1] However, other studies suggest that the risk of blood clots from a PICC line ranges from 3-10%. [11] If a clot forms, the PICC line may be removed and you may be hospitalized for the initiation of anti-coagulation therapy. The risk of clot development is decreased by flushing the PICC line twice daily with saline and heparin and by avoiding vigorous repetitive motion activity of the arm in which the PICC line is placed.

The major side effects of ceftriaxone include rash (in about 1.7% of all cases), diarrhea (2.7%), changes in liver function (approximately 3%) and gallstones (less than 1%). Ceftriaxone is related to penicillin and a small percentage of patients with penicillin allergies will have allergic reactions to ceftriaxone (5-8%). Some patients when treated with ceftriaxone develop pain that seems like gall bladder disease. This usually goes away after the medication is stopped. However, some patients using ceftriaxone have had their gallbladders removed. Some doctors

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believe that this risk can be reduced by taking actigal with the ceftriaxone. Other antibiotics commonly used to treat Lyme disease are not known to have gallbladder side effects. Monitoring your blood may help detect the development of any liver or gall bladder problems.

To my knowledge, I am not allergic to any antibiotics except those listed below:

_____
_____
_____
___

I am aware of the additional risks involved in IV treatment and consent to the use of IV treatments IF my doctor and I agree at a later date to use them.

Signature:_____

Date: _____

Print Name _____

INFORMED LYME CONSENT INCLUDES READING THE PRESIDENT OF THE INTERNATIONAL INFECTION SOCIETY OF AMERICA WHO BELIEVES THE BEST LYME RESEARCH SHOWS:

Lyme is cured in 14-30 Days of antibiotics.

Most Infection Specialists feel Lyme is over diagnosed and is cured in 14-30 Days.

The Use of Antibiotics over 30 days is dangerous and makes super resistant bacteria.

He does not mention much about co-infections, including Bartonella, so my impression is they are not a significant or common issue.

"Chronic" Lyme really not exist at all.

ILADS members treat longer with antibiotics just to get rich.

****SIGN_____DATE_____**

****PRINT_____**

If you or any relative feels this doctor's writing in the article above is right, we would ask that you get treatment from a member of IDSA, and we do not see any reason to be involved in Dr. Schaller's practice.

He does not like treating controversial medical illnesses, and this controversy is one he does not wish to be involved in. **If any relative or if you yourself feel that this position offered above is correct**, then *Dr. Schaller is really the wrong physician for you and you can find plenty of others who will treat you briefly and who will probably take your insurance*. The Quest or LabCorp Lyme basic tests, the Elisa or Western Blot, which they will run will be covered by most insurance companies.

Please do **not** be “sold” by Dr. Schaller. Feel free to get Lyme treatment elsewhere and go to other infection physicians.

SIGN_____
DATE_____

RISK OF PROLONGED ANTIBIOTIC USE

The largest infectious disease society in the US feels Lyme is fully treated in 30 days. This generally is felt to include severe and chronic cases. Therefore, if at any time you want to stop your antibiotics at ANY TIME, feel free to do so. I am willing to treat individuals who feel certain they want longer treatment and it is the best option for them, but only after you have read at least 5 sources of information that ***oppose long treatment with antibiotics for over 30 days.*** **We request strongly you read three articles about any antibiotic you use. Look closely at the side effects and interactions. The Internet offers many useful sites such as:**

1) http://www.drugs.com/

2) http://www.rxlist.com/

3) http://www.mayoclinic.com/health/drug-information/DrugHerbIndex

4) http://www.nlm.nih.gov/medlineplus/druginformation.html

******Make Sure You Learn FULLY About Any Antibiotic You Take.**

****SIGN_____**

****PRINT_____ DATE_____**

Dr. Schaller does not use all of these medications.

All antibiotics cause risk of overgrowth by bad bacteria. One very dangerous and possibly deadly bacteria is C. Diff that causes diarrhea when good bacteria are killed.

Generally, breastfeeding is not recommended while taking antibiotics because of risk of alteration to infant's intestinal good bacteria and hide a serious infection in the infant. Excessive or inappropriate use may promote growth of resistant and dangerous bacteria.

Penicillins: Hypersensitivity may be common, and cross allergenicity with cephalosporins has been reported. Penicillins are classed as category B during **pregnancy**.

Cephalopsorins: Several cephalopsorins and related compounds have been associated with seizures. Cefmetazole, cefoperazone, cefotetan and **ceftriaxone (Ceftin) may be associated with a fall in prothrombin activity and coagulation abnormalities. Pseudomembranous colitis has been reported with cephalosporins and other broad-spectrum antibiotics**. Some drugs in this class may cause renal toxicity. Pregnancy category B.

Fluroquinolones: Lomefloxacin has been associated with increased **photosensitivity**. All drugs in this class have been associated with convulsions. Pregnancy category C.

Tetracyclines: Demeclocycline may cause increased photosensitivity. Minocycline may cause **dizziness**. Do not use tetracyclines in children under the age of eight, and specifically avoid during periods of tooth development. Oral tetracyclines bind to anions such as calcium and iron. **Although doxycycline and minocycline may be taken**

with meals, patients must be advised to take other tetracycline antibiotics on an empty stomach, and not to take the drugs with milk or other calcium-rich foods. Expired tetracycline should never be administered. Pregnancy category D. Use during pregnancy may cause alterations in bone development.

Macrolides: Erythromycin may aggravate the weakness of patients with **myasthenia gravis**. **Azithromycin (Zithromax) has, rarely, been associated with allergic reactions, including angioedema, anaphylaxis, and dermatologic reactions, including Stevens-Johnson syndrome and toxic epidermal necrolysis**. These later serious skin reactions can be deadly. Oral erythromycin may be highly irritating to the stomach and when given by injection may cause severe phlebitis. These drugs should be used with caution in patients with liver dysfunction. Pregnancy category B: Azithromycin, erythromycin. Pregnancy category C: **Clarithromycin (Biaxin)**, dirithromycin, troleandomycin. On example of a newer example of this group is: Ketek. **According to the FDA, three people have suffered severe liver damage as a result of taking the antibiotic Ketek (telithromycin).** This antibiotic medication is commonly used to treat

pneumonia, bronchitis, and sinus infections. The FDA states they are currently evaluating Ketek's safety to determine the frequency of liver problems associated with the drug, manufactured by Sanofi-Aventis. It also cannot be used in people with heart trouble and requires an EKG before use. Drivers may want to take most of their Ketek at night after coming home to avoid a DUI from blurred vision.

Aminoglycosides: This class of drugs causes kidney and **ototoxicity**. These problems can occur even with normal doses. Dosing should be based on renal function, with periodic testing of both kidney function and hearing. Pregnancy category D.

Current Categories for Drug Use in Pregnancy

CategoryDescriptionA Adequate, well-controlled studies in pregnant women have not shown an increased risk of fetal abnormalities.B

Animal studies have revealed no evidence of

harm to the fetus, however, there are no adequate and well-controlled studies in pregnant women.

or

Animal studies have shown an adverse effect, but adequate and well-controlled studies in pregnant women have failed to demonstrate a risk to the fetus. C

Animal studies have shown an adverse effect and there are no adequate and well-controlled studies in pregnant women.

or

No animal studies have been conducted and there are no adequate and well-controlled studies in pregnant women.DStudies, adequate well-controlled or observational, in pregnant women have demonstrated a risk to the fetus. However, the benefits of therapy may outweigh the potential risk.X

Studies, adequate well-controlled or observational, in animals or pregnant women have demonstrated positive evidence of fetal

abnormalities. The use of the product is contraindicated in women who are or may become pregnant.

The misuse of antibiotics leads to strains of bacteria developing resistance to drugs.

Frequent and inappropriate use of antibiotics selects for strains of bacteria that can resist treatment. This is called **bacterial resistance**. These resistant bacteria require higher doses of medicine or stronger antibiotics to treat. Doctors have even found bacteria that are resistant to some of the most powerful antibiotics available today.

Dr. Hudson tells the tale to illustrate how microbes that scientists once thought were easily eliminated by antibiotics can still thrive in the body. His findings and those of other researchers raise disturbing questions about the behavior of microbes in the human body and how they should be treated.

For example, Hudson has found that quiescent varieties of Chlamydia may play a role in chronic ailments not traditionally thought to be related to this infectious agent.

In the early 1990s, he found two types of chlamydia—Chlamydia trachomatis and Chlamydia pneumonia—in the joint tissue of patients with inflammatory arthritis. More famously, in 1996, he began fishing C. pneumonia out of the brain cells of Alzheimer's victims. Since then, other researchers have made headlines after reporting the genetic fingerprints of C. pneumonia, as well as several kinds of common mouth bacteria, in the arterial plaque of heart attack patients. Hidden infections are now thought to be the basis of still other stubbornly elusive ills like chronic fatigue syndrome, Gulf War syndrome, multiple sclerosis, lupus, Parkinson's disease, and types of cancer.

A study published in the *Journal of the American Medical Association* (JAMA)* provides evidence that use of antibiotics is associated with an increased risk of breast cancer. The authors concluded that the more antibiotics the women in the study used, the higher their risk of breast cancer. The results of this study do not mean that antibiotics cause breast cancer. "These results only show that there is an association between the two," explained co-author Stephen H. Taplin, M.D.

A Link Between Antibiotics and Asthma

Is there something **parents** can do to change whether their children will get **asthma**? Babies who received a course of **antibiotics** during the **first six months of life** are 2.5 times more likely than their peers to have developed asthma by **age 7**, according to a Henry Ford Health System study. And babies who took even one round broad-spectrum antibiotics were 8.9 times more likely to acquire asthma. The study, presented at the European Respiratory Society's September 2003 annual conference, and reported by Reuters Health, also demonstrated a link between infants antibiotic use and other **allergic** diseases such as **eczema**. The link was clear -- but which causes which? Are children who have allergic diseases more likely to get sick during the first six months? Or do the antibiotics increase the risk of allergic disease? To some extent both may be true. But recent controlled studies showing that giving **probiotics** (active cultures of **beneficial bacteria**) **lowers** the risk of allergic disease suggest that these data point to antibiotics actively raising the risk. What's a parent to do? If someone suggests antibiotics for your child, ask if there is a **safe way to treat the illness without them**. If antibiotics are important to give, ask for the most narrow-spectrum choice that would work (antibiotics that kill the desired **bacteria**, and as few of the bystander bacteria as practical). Whenever a child

needs antibiotics, I suggest giving probiotics to replenish the diversity of beneficial bacteria.
SOURCE:http://www.drgreene.com/21_1694.html

I can neither oppose nor support these comments from a wide range of locations.

If you are given probiotics, please understand most physicians do not do this and no promises of safety or purity or protection can be offered.
*****SIGN_____**
*****DATE_____**

I AM ONLY OFFERING FOR YOUR EXTENSIVE AND FULL INFORMED CONSENT.

Pick one choice please:

1) I WILL TAKE ANTIBIOTICS 30 DAYS___

2) I WILL TAKE UNTIL MY CLINICAL TROUBLES IMPROVE___

3) I WILL START THEM AND THINK OVER THEIR DURATION ___

4) I DO NOT WANT ANY ANTIBIOTICS____

Print name_____
Sign_____ Date_____

*****SIDE EFFECTS AND INTERACTIONS*****

I understand I am being asked to read *three* articles on any antibiotic I am prescribed TO LEARN SIDE EFFECTS, INTERACTIONS, RISKS AND POSSIBLE BENEFITS.

_____

The Internet offers many useful sites such as:

1) **http://www.drugs.com/**

2) **http://www.rxlist.com/**

3) **http://www.mayoclinic.com/health/drug-information/DrugHerbIndex**

4) **http://www.nlm.nih.gov/medlineplus/druginformation.html**

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Lyme Disease: Myths and Reality

To the Editor:

"More Awareness, More Cases" (June 4) provides beneficial information regarding reasons for the yearly fluctuations in Lyme disease cases, basic information on disease transmission and patient symptoms, and successful prevention efforts in New Jersey. However, the article also perpetuates common but inaccurate myths about Lyme disease.

One of these myths concerns Lyme disease testing. When used appropriately, Lyme disease testing is actually highly accurate. The estimate of 50 percent accuracy is true only when testing is inappropriately used early in the disease course during the rash phase of the illness where laboratory testing is not recommended or needed for diagnosis.

In addition, the article presents one clinician's observation that chronic Lyme disease accounts for 15 to 20 percent of cases. However, there is no credible scientific evidence for the persistence of symptomatic Borrelia burgdorferi infection after antibiotic treatment.

The unfortunate truth is that more than 50 percent of patients who are labeled with the diagnosis of "chronic Lyme disease" have never actually had any form of Lyme disease. In one study, published in Annals of Internal Medicine, the majority of these patients actually had other treatable conditions that had gone undiagnosed and untreated.

If readers want reliable information on Lyme disease, Web sites such as medlineplus.gov **may be used.**

Gary P. Wormser, M.D., Raymond J. Dattwyler, M.D.
Valhalla
The writers are chairmen of, respectively, the Division of Infectious Diseases and the Division of Allergy, Rheumatology and Clinical Immunology, New York Medical College.

These physicians report that routine Lyme testing is entirely accurate 4-6 weeks after a tick bite. They say there is no such thing as chronic Lyme disease or persistent Lyme disease. Further, they believe ½ of those who are told they have “Chronic Lyme disease” never had any Lyme.

Dr. Schaller provides as much conservative material as possible. *He does not want to diagnose or treat Lyme in*

any form. Feel free to pursue other physicians and to have them treat your medical or psychiatric trouble. You can find a physician that reflects Dr. Wormser's views at IDSA or the Infectious Disease Society of America. Most infection specialists agree with his positions.

http://www.nytimes.com/2006/06/09/opinion/nyregionopinions/l11west.html?_r=1&oref=slogin

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DATE_____

Lyme Disease Treatment Is Source Of Contention

The July 31 Other Opinion article by Dr. Raphael B. Stricker is filled with inaccuracies and misleading information that can only contribute to the public's misunderstanding and unfounded fears about Lyme disease.

For example, the "small group of scientists" whom Dr. Stricker states are impeding effective treatments for the disease is in fact

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the 8,000-member Infectious Diseases Society of America, the nation's largest professional association of infectious disease experts. Ironically, it is Dr. Stricker who represents "a small group of scientists." The International Lyme and Associated Diseases Society, of which he is president, has approximately 200 members, many of whom have a vested interest in promulgating long-term treatment for Lyme disease because they make their living off such treatment. Here's what else Courant readers need to know:

The IDSA Clinical Practice Guidelines for Lyme disease, which Dr. Stricker dismisses, were developed by a panel of world-renowned experts in Lyme disease, doctors who are researchers and who regularly treat Lyme disease patients. The guidelines were created based on stringent rules of evidence-based medicine, which includes a thorough and objective review of published scientific research. This comprehensive review also included research submitted by members of Dr. Sticker's organization. The guidelines undergo a rigorous peer review and must ultimately be approved by the IDSA board of directors, who are all infectious-disease specialists and leaders in the field.

Despite claims that many people suffer from "chronic Lyme disease," a thorough review of published research reveals no credible scientific evidence that the disease persists after appropriate antibiotic treatment indicated in the IDSA guidelines.

There is valid scientific evidence documenting that more than 50

percent of patients diagnosed with "chronic Lyme disease" never actually had any form of Lyme disease. In one published scientific study, the majority of such patients were found to have conditions that, because of the misdiagnosis of chronic Lyme disease, were not properly diagnosed or treated.

In fact, Lyme disease is a tick-borne bacterial infection. In the vast majority of cases (95 percent to 98 percent), the disease is successfully treated with a two-week course of oral antibiotics. A small number of patients may require longer treatment of up to one month. These facts are supported overwhelmingly by the most current published studies.

There are a very few patients treated for Lyme disease who continue to have symptoms after the recommended treatment. The long-term antibiotic therapy advocated by Dr. Stricker and his colleagues can endanger these individuals, cause needless suffering and may promote the development of drug-resistant "superbugs."

We encourage anyone who is diagnosed with so-called chronic Lyme disease to get a second opinion. We strongly advise that people with Lyme disease avoid long-term antibiotic treatment, which is ineffective, costly and potentially very dangerous.

IDSA members have been at the forefront of preventing and treating Lyme disease. Our singular mission is to find out what is best for patients. We believe that our guidelines represent the best

that science has to offer.

For reliable, scientific-based information on Lyme disease, we would direct your readers to sources such as the websites of the American College of Physicians (www.acponline.org) and the U.S. Centers for Disease Control and Prevention (www.cdc.gov), as well as that of IDSA (www.idsociety.org).

Martin J. Blaser, M.D.
President
Infectious Diseases Society of America

If you or any relative feels Dr. Blaser's writing in the article above is right, we would ask that you get treatment from a member of IDSA, and we do not see any reason to be involved in Dr. Schaller's practice.

He does not like treating controversial medical illnesses, and this controversy is one he does not wish to be involved in. If any relative or if you yourself feel that this position offered above is correct, then *Dr. Schaller is really the wrong physician for you and you can find plenty of others who will treat you briefly and who will probably take your insurance*. The Quest or LabCorp Lyme basic tests, the Elisa or Western Blot, which they will run will be covered by most insurance companies.

Please do not be "sold" by Dr. Schaller. Feel free to get Lyme treatment elsewhere and go to other infection physicians.

Sign_____
Date_____

Print Name _____

ILADS Demands Retraction of New IDSA Guidelines for Treatment of Lyme Disease

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To: National Desk, Health Reporter

Contact: Raphael Stricker, 415-823-4007 or Richard Horowitz, 845-229-8977 or Barbara Buchman, 301-263-1080, all of International Lyme and Associated Diseases Society

WASHINGTON, Oct. 27 /U.S. Newswire/ -- Recently published Lyme disease guidelines from the Infectious Diseases Society of America (IDSA) "threaten to harm patients and patient care," writes Dr. Raphael Stricker,

president of the International Lyme and Associated Diseases Society (ILADS) in an angry and highly unusual letter to Dr. Sherwood Gorbach, editor of 'Clinical Infectious Diseases.' The letter calls for retraction of the article, "The clinical assessment, treatment, and prevention of Lyme disease, human granulocytic anaplasmosis, and babesiosis: Clinical practice guidelines by the Infectious Diseases Society of America."

Authors used exclusionary data selection, writes Dr. Stricker, which substantially biased the resulting diagnosis and treatment recommendations and ignored opposing or dissenting views.

Lyme disease is a major public health problem caused by a complex bacterial agent carried and spread by ticks. Lyme bacteria have the ability to evade immune destruction, entrench themselves deep within tissues and migrate throughout the body causing a multi-system illness. There is no single reliable diagnostic laboratory test and there is no universally accepted treatment for Lyme disease.

In a point-by-point analysis of the guidelines, Dr. Stricker states IDSA's position reflects a biased view of the disease that's either unsubstantiated or refuted by existing peer-reviewed medical literature. "The IDSA's 'one-size fits all' approach to Lyme diagnosis and treatment will result in

misdiagnosis and denial of care to thousands of patients annually," writes Dr. Stricker in the letter to Dr. Gorbach, "creating a public health burden as...patients become chronically ill and disabled."

"We believe the same principles of scientific integrity that apply to medical research should also apply to practice guidelines," states Dr. Stricker. The IDSA authors deliberately excluded divergent points of view, he adds. As a result, ten of IDSA's central recommendations are based solely on opinion, considered the weakest level of evidence in science circles.

ILADS demands immediate retraction of the guidelines article and subsequent CDC notification; creation of a new Lyme disease guidelines committee comprised of diverse points of view to reformulate the guidelines; and then submission of revised guidelines for outside peer review to a medical journal independent of IDSA.

Dr. Stricker's letter was widely distributed to CDC, NIH, FDA and leading national physician organizations.

ILADS (**http://www.ilads.org**) is a multi-specialty medical society comprised of all sub-specialists who treat Lyme disease, including infectious disease specialists, neurologists, rheumatologists, psychiatrists,

endocrinologists and internal medicine physicians. President Dr. Raphael B. Stricker is medical director of Union Square Medical Associates in San Francisco.

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The Attorney General of Connecticut Goes After IDSA

"AG Takes on MDs Over Lyme Disease "

"Blumenthal subpoenas national doctors group over treatment guidelines"

"In an unprecedented move, the state attorney general has waded into the controversy surrounding the diagnosis and treatment of Lyme disease by launching an antitrust probe into the practices of a national doctors' group...."

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http://www.conntact.com/article_page.lasso?id=40479

"This isn't theoretical," Blumenthal said. "It will come down to a matter of dollars and cents."
http://www.newstimeslive.com/news/story.php?id=1023559

Lyme Disease Guidelines Focus Of Antitrust Probe

By ELIZABETH HAMILTON
Courant Staff Writer

November 17 2006

Attorney General Richard Blumenthal said Thursday that he has launched an investigation into whether the Infectious Diseases Society of America has violated antitrust laws in setting new guidelines for diagnosing and treating Lyme disease.

The investigation, the latest chapter in a bitter controversy over the use of long-term antibiotics to treat Lyme disease, comes as the state Medical Examining Board wraps up a hearing for a New Haven doctor accused of violating standards of care in treating two young patients he diagnosed with the disease.

The attorney general's investigation, which could result in a lawsuit, revolves around guidelines set last month by the IDSA and since adopted by the Centers for Disease Control and Prevention. Blumenthal and others say the guidelines are being used by insurance companies to determine - and sometimes deny - coverage to people asking for certain types of treatment for Lyme disease.

The IDSA is an influential organization composed of 8,000 members with training in infectious diseases. A panel of physicians and scientists at the organization set the latest guidelines, which critics say are predicated on the belief that Lyme disease can be diagnosed by confirming specific, objective symptoms, such as the bull's-eye rash, and that it can be cured with a limited course of antibiotics.

But a smaller group of doctors and academics believe the medical establishment has it wrong

and is effectively denying patients desperately needed treatment by setting such restrictive guidelines.

Blumenthal said the disagreement over Lyme disease, and the effect the new guidelines are having on Lyme disease patients, is what prompted him to look into the IDSA's conclusions and how they were reached.

"These guidelines were set by a panel that essentially locked out competing points of view," Blumenthal said. "Presumably, the IDSA is a non-profit making organization, but such organizations can still be used for anti-competitive purposes."

Blumenthal also said his office has received numerous complaints from both patients and doctors about the guidelines.

"One of the common complaints we've received relates to denials of insurance coverage, that XYZ insurer won't cover this form of treatment because the guidelines make no provisions for it," he said. "It's a very chilling economic effect."

Diana Olson, the director of communications for the IDSA in Virginia, did not return a call for comment Thursday.

http://www.courant.com/news/local/hc-ctlyme1117.artnov17,0,1670696.story

GUIDELINES FROM THE INFECTIOUS DISEASES SOCIETY OF AMERICA
Practice Guidelines for the Treatment of Lyme Disease

Gary P. Wormser,1 Robert B. Nadelman,1 Raymond J. Dattwyler,2 David T. Dennis,6 Eugene D. Shapiro,7 Allen C. Steere,9 Thomas J. Rush,5 Daniel W. Rahn,10 Patricia K. Coyle,3 David H. Persing,11 Durland Fish,8 and Benjamin J. Luft4

Executive Summary

Tick bites and prophylaxis. The best currently available method for preventing infection with *Borrelia burgdorferi* is to avoid vector tick exposure. If exposure to *Ixodes scapularis* or *Ixodes pacificus* ticks is unavoidable, measures recommended to reduce the risk of infection include using both protective clothing and tick repellents, checking the entire body for ticks daily, and promptly removing attached ticks, before transmission of *B. burgdorferi* can occur (A-III [see tables 1 and 2 for recommendation categories, indicated in parentheses throughout this text]).

Routine use of either antimicrobial prophylaxis (E-I) or serological tests (D-III) after a tick bite is not recommended.

Some experts recommend antibiotic therapy for patients bitten by *I. scapularis* ticks that are estimated to have been attached for 148 h (on the basis of the degree of engorgement of the tick with blood), in conjunction with epidemiological information regarding the prevalence of tick-transmitted infection (C-III). However, accurate determinations of species of tick and

degree of engorgement are not routinely possible, and data are
insufficient to demonstrate efficacy of antimicrobial therapy in
this setting.

Persons who remove attached ticks should be monitored closely for signs and symptoms of tick-borne diseases for up to 30 days and specifically for the occurrence of a skin lesion at the site of the tick bite (which may suggest Lyme disease) or a temperature 1387C (which may suggest human granulo-

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cytic ehrlichiosis [HGE] or babesiosis). Persons who develop a
skin lesion or other illness within 1 month after removing an attached tick should promptly seek medical attention for assessment
of the possibility of having acquired a tick-borne disease (A-II).

Health care practitioners, particularly those in areas where Lyme disease is endemic, should become familiar with its clinical
manifestations, recommended practices for testing for it, and therapy for the disease, as well as for HGE and babesiosis
(A-III).

Testing of ticks for tick-borne infectious organisms is not

recommended, except in research studies (D-III).
Prior vaccination with the recently licensed recombinant outer-surface protein A (OspA) vaccine preparation reduces the
risk of developing Lyme disease associated with tick bites but
should not alter the above recommendations (A-I).
Early Lyme disease. Administration of doxycycline (100 mg twice daily) or amoxicillin (500 mg 3 times daily) for 14–21
days is recommended for treatment of early localized or early
disseminated Lyme disease associated with erythema migrans,
in the absence of neurological involvement or third-degree atrioventricular heart block (A-I). In prospective studies, these
agents have been shown to be effective in treating erythema migrans and associated symptoms. Doxycycline has the advantage
of being efficacious for treatment of HGE, which may occur simultaneously with early Lyme disease. Doxycycline is
relatively contraindicated during pregnancy or lactation and for children aged !8 years.
Because of its higher cost, cefuroxime axetil (500 mg orally twice daily), which is as effective as doxycycline in the treatment
of erythema migrans (A-I), should be reserved as an alternative

agent for those patients who can take neither doxycycline nor
amoxicillin. For children, we recommend amoxicillin at a dosS2
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Table 1. Categories indicating the strength of each recommendation
for or against use.

Category	Definition
A	Good evidence to support a recommendation for use
B	Moderate evidence to support a recommendation for use
C	Poor evidence to support a recommendation for or against use
D	Moderate evidence to support a recommendation against use
E	Good evidence to support a recommendation against use

NOTE. Table is adapted from [1].

Table 2. Grades indicating the quality of evidence on which recommendations
are based.

Grade	Definition
I	Evidence from at least 1 properly randomized, controlled trial
II	Evidence from at least 1 well-designed clinical trial without randomization,
from cohort or case-controlled analytic studies
(preferably from 11 center), from multiple time-series studies, or
from dramatic results of uncontrolled experiments

III Evidence from opinions of respected authorities that is based on clinical experience, descriptive studies, or reports of expert committees

age of 50 mg/kg/d, divided into 3 doses per day (maximum, 500 mg/dose), or doxycycline (for those aged >8 years) at a dosage of 1–2 mg/kg twice per day (maximum, 100 mg/dose) (B-II). Cefuroxime axetil, at a dosage of 30 mg/kg/d, divided into 2 doses daily (maximum, 500 mg/dose), is an acceptable alternative agent (B-III).

Macrolide antibiotics are not recommended as first-line therapy for early Lyme disease (E-I). When used, they should be reserved for patients who are intolerant of amoxicillin, doxycycline, and cefuroxime axetil. Possible regimens for adults are as follows: azithromycin, 500 mg orally daily for 7–10 days; erythromycin, 500 mg orally 4 times daily for 14–21 days; and clarithromycin, 500 mg orally twice daily for 14–21 days. Possible dosages for children are the following: azithromycin, 10 mg/kg/d (maximum, 500 mg/d); erythromycin, 12.5 mg/kg 4 times daily (maximum, 500 mg/dose); and clarithromycin, 7.5 mg/kg twice daily (maximum, 500 mg/dose). Patients treated with macrolides should be closely followed.

Ceftriaxone (2 g iv daily), although effective, is not superior

to oral agents and is not recommended as a first-line agent for
treatment of Lyme disease in the absence of neurological involvement
or third-degree atrioventricular heart block (E-I).
The use of ceftriaxone (2 g once daily iv for 14–28 days) in early Lyme disease is recommended for acute neurological disease
manifested by meningitis or radiculopathy (B-II). Intravenous penicillin G at a dosage of 18–24 million units daily,
divided into doses given every 4 h (for patients with normal renal function), may be a satisfactory alternative (B-II). Cefotaxime
(2 g iv every 8 h) may also be a satisfactory alternative (B-II). For adult patients who are intolerant of both penicillin and cephalosporins, doxycycline (200–400 mg/d) in 2 divided doses given orally (or iv if the patient is unable to take oral medications) for 14–28 days may be adequate (B-II).
For children, we recommend ceftriaxone (75–100 mg/kg/d) in a single daily iv dose (maximum, 2 g) (B-II) or cefotaxime (150–200 mg/kg/d) divided into 3 or 4 iv doses (maximum, 6 g/d) (B-III) for 14–28 days. An alternative is iv penicillin G (200,000–400,000 units/kg/d; maximum, 18–24 million units/d)
divided into doses given every 4 h for those with normal renal
function (B-II).
Patients with first- or second-degree atrioventricular heart block associated with early Lyme disease should be treated with

the same antimicrobial regimens as patients with erythema migrans without carditis (see paragraphs 1 and 2 of the recommendations in this section, above) (B-III). We recommend that patients with third-degree atrioventricular heart block be treated with parenteral antibiotics such as ceftriaxone (see paragraphs 5 and 6 of the recommendations in this section, above) in the hospital, although there are no clinical trials to support this recommendation (B-III). A temporary pacemaker may also be required.

Although antibiotic treatment does not hasten the resolution of seventh-cranial-nerve palsy associated with *B. burgdorferi* infection, antibiotics should be given to prevent further sequelae (B-II). There was disagreement among panel members on the neurological evaluation of patients with seventh-cranial-nerve palsy. Some members perform a CSF examination on all patients with seventh-cranial-nerve palsy, whereas others reserve lumbar puncture for patients for whom there is strong clinical suspicion of CNS involvement (e.g., severe headache or nuchal rigidity). Patients whose CSF examinations yield normal findings

may be treated with the same regimens used for patients with erythema migrans (B-III), whereas patients for whom there is clinical and laboratory evidence of CNS involvement should be treated with regimens effective against meningitis (see paragraphs 5 and 6 of the recommendations in this section, above) (B-II).

Treatment for pregnant patients can be identical to that for nonpregnant patients with the same disease manifestation, except that tetracyclines should be avoided (B-III).

Lyme arthritis. Lyme arthritis usually can be treated successfully with antimicrobial agents administered orally or intravenously.

Administration of doxycycline (100 mg twice daily orally) or amoxicillin (500 mg 3 times daily), in each instance for 28 days, is recommended for patients without clinically evident neurological disease (B-II). For children, we recommend administration of doxycycline (1–2 mg/kg twice per day; maximum, 100 mg/dose), which can be given to patients aged >8 years, or amoxicillin (50 mg/kg/d, divided into 3 doses per day; maximum, 500 mg/dose) for 28 days (B-II).

Oral therapy is easier to administer than iv antibiotics, is

associated with fewer serious complications, and is considerably
less expensive. Its disadvantage is that some patients treated
with oral agents have subsequently manifested overt neuroborreliosis,
which may require iv therapy for successful treatCID 2000;31 (Suppl 1) IDSA Guidelines for Lyme Disease S3
ment. Further controlled trials are needed to compare oral with
iv therapy.

Neurological evaluation, including lumbar puncture, should be done for patients if there is a strong clinical suspicion of neurological involvement. Patients with both arthritis and objective
evidence of neurological disease should receive iv ceftriaxone
(2 g once daily for 14–28 days) (A-II). Alternative
therapies include iv cefotaxime (2 g iv every 8 h) (B-III) or iv penicillin G (18–24 million units daily, divided into doses given
every 4 h for patients with normal renal function) (B-II). Because
of low blood levels, the long-acting benzathine preparation of penicillin is not recommended (D-III). For children,
we recommend administration of ceftriaxone (75–100 mg/kg/d
in a single daily iv dose; maximum, 2 g) (B-III) or cefotaxime (150–200 mg/kg/d divided into 3 or 4 iv doses; maximum, 6 g/

d) (B-III) for 14–28 days. An alternative is iv penicillin G (200,000–400,000 units/kg/d; maximum, 18–24 million units/d), divided into doses given every 4 h for those with normal renal function (B-III).

For patients who have persistent or recurrent joint swelling after recommended courses of antibiotic therapy, we recommend repeat treatment with another 4-week course of oral antibiotics or with a 2- to 4-week course of iv ceftriaxone (B-III). Clinicians should consider waiting several months before initiating repeat treatment with antimicrobial agents because of the anticipated slow resolution of inflammation after treatment.

If patients have persistent arthritis despite 2 courses of oral therapy or one course of iv therapy, symptomatic treatment with nonsteroidal anti-inflammatory agents is recommended; intra-articular steroids may also be of benefit (B-III). If persistent synovitis is associated with significant pain or if it limits function, arthroscopic synovectomy can reduce the period of joint inflammation (B-II).

Late neuroborreliosis affecting the CNS or peripheral nervous system. For patients with late neurological disease affecting the CNS or peripheral nervous system, treatment with ceftriaxone

(2 g once a day iv for 2–4 weeks) is recommended (B-II). Alternative parenteral therapy may include administration of cefotaxime (2 g iv every 8 h) (B-II) or iv penicillin G (18–24 million units daily, divided into doses given every 4 h for patients with normal renal function) (B-II). Response to treatment is usually slow and may be incomplete. However, unless relapse is shown by reliable objective measures, repeat treatment is not recommended. For children, a 14–28-day course of treatment with ceftriaxone (75–100 mg/kg/d in a single daily iv dose; maximum, 2 g) is recommended (B-II). An alternative is cefotaxime (150–200 mg/kg/d iv, divided into 3 or 4 doses; maximum, 6 g/d) (B-II). Another alternative is iv penicillin G (200,000–400,000 units/kg/d, divided into doses given every 4 h for those with normal renal function; maximum, 18–24 million units/d) (B-II).

Chronic Lyme disease or post–Lyme disease syndrome. After an episode of Lyme disease that is treated appropriately, some persons have a variety of subjective complaints (such as myalgia, arthralgia, or fatigue). Some of these patients have been classified

as having “chronic Lyme disease” or “post–Lyme disease syndrome,”
which are poorly defined entities. These patients appear to be a heterogeneous group. Although European patients rarely
have been reported to have residual infection with *B. burgdorferi,*
this has yet to be convincingly demonstrated either in a large
series of appropriately treated European patients or in a study
of North American patients.
Randomized controlled studies of treatment of patients who remain unwell after standard courses of antibiotic therapy for
Lyme disease are in progress. ***To date, there are no convincing published data that repeated or prolonged courses of either oral or iv antimicrobial therapy are effective for such patients. The consensus of the Infectious Diseases Society of America (IDSA)***

***expert-panel members is that there is insufficient evidence to regard “chronic Lyme disease” as a separate diagnostic entity*.**

I READ PARAGRAPH ABOVE FROM IDSA

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Objective

The objective of these practice guidelines is to provide clinicians and other health care practitioners with recommendations for management of cases in which either Lyme disease has been diagnosed or the patient was bitten by an *Ixodes* tick in North America (tables 1 and 2) [1]. Lyme disease is endemic in several regions of the United States, particularly areas of the Northeast, Upper Midwest, and Northwest [2]. It is the most

frequent vector-borne disease in the United States. Adults and children of both sexes can be affected. These patients are evaluated and treated by general practitioners, pediatricians, and internists, as well as by infectious disease specialists, dermatologists, rheumatologists, neurologists, orthopedists, obstetricians, and ophthalmologists. Because the genospecies of *B. burgdorferi* that cause Lyme disease in North America are different from those that cause Lyme borreliosis in Eurasia, recommendations were based, whenever possible, on studies conducted in the United States.

In the treatment of this disease, as in all infectious diseases, basic medical and scientific principles should be considered. In selecting an antibiotic, there should be evidence of activity in vitro, evidence of penetration into the infected sites, and clinical studies to support the treatment regimen. The reader is referred to other sources for information on diagnostic aspects of Lyme disease [3–9].

Prevention of Tick Bites

The best currently available method for preventing infection with *B. burgdorferi* and other *Ixodes*-transmitted infections is

to avoid tick-infested areas [10]. If exposure to *I. scapularis* or
I. pacificus ticks is unavoidable, a number of measures may help to decrease the risk that ticks will attach and subsequently
transmit infection. The use of protective clothing (shirt tucked
into pants and pants tucked into socks) may interfere with
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attachment by ticks by increasing the time required for ticks to find exposed skin, thus facilitating their recognition and removal.
By wearing light-colored clothing (to provide a background with which the tick contrasts), persons in areas of endemicity may also be more likely to see (and remove) ticks
before they have attached.
Daily inspections of the entire body to locate (and remove) ticks also provide an opportunity to prevent transmission of tick-borne infections [11, 12]. Attached ticks should be removed
promptly with fine-toothed forceps, if possible [13]. Tick and insect repellents applied to the skin and clothing provide additional
protection [10, 14, 15].

Tick Bites and Prophylaxis

Primary Management Options

For patients who remove attached ticks, we considered the following management options: (1) treating all such persons; (2) treating only persons believed to be at high risk (e.g., those

removing a nymphal or adult vector tick [*I. scapularis* or *I. pacificus*] after 48 h of attachment); (3) treating only persons who develop erythema migrans or other clinical signs and symptoms of tick-borne infection; and (4) treating all persons who seroconvert from negativity to positivity (optimally with a 4-fold increase in titer) for serum antibodies to *B. burgdorferi* (acute and follow-up blood specimens from all persons who are bitten would need to be collected and tested for antibodies in paired specimens).

Outcome

The panel weighed both the risks and the consequences of developing Lyme disease (including the risk of late complications) for persons bitten by vector ticks *(I. scapularis* or *I. pacificus)* against the cost and adverse effects of prophylactic antimicrobials. The effect of the different strategies on quality of life was considered. In addition, we considered the effect of the recent licensing of a recombinant OspA vaccine for prevention of Lyme disease [16]. The principal desired outcome is prevention of Lyme disease. Another desired outcome is the prevention of other *Ixodes*-borne illnesses, including babesiosis

and HGE. Concurrent infection and disease with these organisms
have been described [17–19].

Evidence

Option 1: treating with antimicrobials all persons who remove
vector ticks (I. scapularis *or* I. pacificus) *that have become attached.*

Although some practitioners routinely treat patients
that have been bitten by *I. scapularis* [20], several prospective,
randomized double-blind clinical trials involving persons who were bitten by *I. scapularis* ticks and then were treated with placebo, penicillin, tetracycline, or amoxicillin each led to conclusions
that routine antimicrobial prophylaxis is not warranted [21–23]. A meta-analysis of these studies (in which 1600 persons
were enrolled) did not indicate that antimicrobial prophylaxis is effective (pooled OR, 0.0; 95% CI, 0.0–1.5; *P*p.12) [24]. The authors of the meta-analysis estimated that if amoxicillin
rather than doxycycline were used (to enable small children and
pregnant or lactating women to receive prophylaxis), 8 cases of drug-associated rash, including 1 severe life-threatening reaction,
would occur for every 10 cases of early Lyme disease
that were prevented [24].

In addition, 3 cases of minor amoxicillin-related adverse effects
(e.g., diarrhea) would occur for every case of Lyme disease that was prevented. In 2 studies of prophylaxis for tick bites in which adverse effects of the antimicrobials used for prophylaxis
were reported, the risk of acquiring Lyme disease after a tick bite was no different than the risk of developing adverse effects from the prophylactic antibiotics [21, 22].
One cost-effectiveness analysis concluded that a 2-week course of doxycycline is indicated when the probability of infection
with *B. burgdorferi* after a tick bite is >.036 and should be considered when the theoretical probability ranges from .01
to .035 [25]. Many experts, however, disagree with key assumptions
in the model. Furthermore, doxycycline is relatively contraindicated for women who are either pregnant or breastfeeding,
as well as for children aged !8 years.
Some practitioners prescribe a 10-to-14-day course of amoxicillin
for pregnant women who have been bitten by *I. scapularis*, because case reports have suggested that adverse outcomes
for the fetus may be associated with pregnancies complicated by Lyme borreliosis [26]. Increasing data from clinical

and epidemiological studies, however, suggest that favorable outcomes can be expected when pregnant women with Lyme borreliosis are treated with standard antibiotic regimens [27–29].

In addition to *B. burgdorferi,* other potential pathogens may be present in *I. scapularis* ticks [30, 31]. Babesiosis and HGE can occur independently or together with Lyme disease [17, 18, 32]. Administration of doxycycline is effective in the treatment of patients with HGE [33] but is not recommended as therapy for babesiosis. There are no published clinical data regarding the efficacy of prophylaxis with doxycycline against either of these infections.

Option 2: treating with antimicrobials only persons believed to be at high risk (e.g., those who have removed a nymphal or adult vector tick [I. scapularis *or* I. pacificus] *after 48 h of attachment).*

Entomological studies have shown that *B. burgdorferi* is rarely transmitted by *I. scapularis* within the first 48 h of attachment to laboratory animals [11, 12]. This “grace period” is required

for spirochetes to migrate from the gut into the salivary glands
of infected ticks once feeding commences [34]. Thus, ticks that
have been attached for !48 h theoretically cannot transmit *B.*
burgdorferi infection. However, this is not true for HGE or babesiosis, since the organisms that cause these diseases are
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already present in the salivary glands before feeding (D. Fish,
unpublished data, and [35]).
The option of treating selectively persons with “high-risk” tick bites to prevent Lyme disease assumes that the species, stage, degree of engorgement, and infection status of the tick,
as well as the probability of transmission of infection, can be readily ascertained. This is rarely true. Many different tick species
bite humans, and some “ticks” removed from humans are actually spiders, scabs, lice, or dirt and thus pose no risk of Lyme disease [36, 37]. Methods for determining the infection status of ticks removed from patients are experimental and are
not standardized. One study found that patients who removed
partially engorged ticks that were calculated to have been attached
for >72 h were significantly more likely to develop *B.*

burgdorferi infection than were patients who removed ticks that had been attached for an estimated duration of !72 h (*P* p .008) [37]. However, even if the risk of Lyme disease is increased with partially engorged ticks, no study has demonstrated that antimicrobials are effective in reducing the risk of infection after a tick bite.

Option 3: treating with antimicrobials only persons who develop erythema migrans or other clinical manifestations of Lyme disease or other tick-transmitted diseases. The great majority of persons with *B. burgdorferi* infection present with erythema migrans [16, 38–40]. Since primary erythema migrans lesions occur at the site of a tick bite [41–44], a person who removes a tick would be likely to detect and to seek care for a rash that subsequently develops at that location. Patients who develop fever in the absence of erythema migrans after an *Ixodes* tick bite should be evaluated for HGE and/or babesiosis in areas where these infections are endemic [33, 45, 46].

In a placebo-treated population observed prospectively in a

large, multicenter vaccine trial, some volunteers developed serological
evidence of asymptomatic *B. burgdorferi* infection [16]. Whether antibiotic therapy is beneficial for such patients is unknown, a question in need of further study. (See next paragraph
[option 4] for caveats concerning serological diagnosis.)

Option 4: treating with antimicrobials all persons who seroconvert
from negativity to positivity for serum antibodies to B. burgdorferi when acute and follow-up serum samples are tested
simultaneously. **Although assessment of acute- and convalescent-**
phase serologies is a standard means of identifying individuals
with a variety of infectious diseases, the utility of this approach for identifying infection with B. burgdorferi following
a tick bite is unknown. Present serological assays for Lyme
disease have substantial limitations [3–7], and their use is not
recommended for screening of persons lacking objective manifestations
of Lyme disease [3, 4, 6, 7].

Recommendations

The best currently available method for preventing infection with *B. burgdorferi* and other *Ixodes*-transmitted infections is to avoid vector tick exposure. If exposure to *I. scapularis* or *I.*

pacificus ticks is unavoidable, measures recommended to reduce the risk of infection include using both protective clothing and tick repellents, checking the entire body for ticks daily, and promptly removing attached ticks before transmission of *B. burgdorferi* can occur (A-III).

Routine use of either antimicrobial prophylaxis (E-I) or serological tests (D-III) after a tick bite is not recommended.

Some experts recommend antibiotic therapy for patients bitten by *I. scapularis* ticks that are estimated to have been attached for 148 h (on the basis of the degree of engorgement of the tick with blood), in conjunction with epidemiological information regarding the prevalence of tick-transmitted diseases (CIII). However, accurate determinations of tick species and degree of engorgement are not routinely possible, and data are insufficient to demonstrate efficacy of antimicrobials in this setting.

Persons who remove attached ticks should be monitored closely for signs and symptoms of tick-borne diseases for up to 30 days and specifically for the occurrence of a skin lesion at the site of the tick bite (which may suggest Lyme disease) or a temperature 1387C (which may suggest HGE or babesiosis).

Persons who develop a skin lesion or other illness within

1 month after removing an attached tick should promptly seek medical attention for assessment of the possibility of having acquired a tick-borne disease (A-II).

Health care practitioners, particularly those in areas where Lyme disease is endemic, should become familiar with the clinical manifestations of and recommended practices for testing and therapy for Lyme disease, as well as for HGE and babesiosis (A-III).

Testing of ticks for tick-borne infectious organisms is not recommended, except in research studies (D-III).

Prior vaccination with the recently licensed recombinant OspA vaccine preparation reduces the risk of developing Lyme disease associated with tick bites but should not alter the above recommendations (A-I).

Early Lyme Disease

Primary Management Options

We considered the following management options for early Lyme disease: oral antimicrobial therapy for early localized infection (i.e., solitary erythema migrans) and oral versus iv therapy for cases of early disseminated infection (i.e., patients presenting with multiple erythema migrans lesions, carditis, cranial-nerve palsy, meningitis, or acute radiculopathy). Borrelial

lymphocytoma was not addressed because of its rarity in North
America (its primary causative organism, *Borrelia afzelii,* is an
exclusively Eurasian genospecies).
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Outcome

The panel weighed both the risks and the consequences of developing late complications of Lyme disease and the possible
adverse effects of antimicrobial therapy. The desired outcome
is to resolve the symptoms and signs of early Lyme disease and
to prevent late complications.

Evidence

At least 7 randomized prospective trials have addressed the treatment of early Lyme disease in the United States [47–53].
All studies used erythema migrans as the disease-defining criterion.
Six studies recruited patients with either localized or disseminated early Lyme disease [47–52], whereas 1 study required
disseminated early disease for enrollment [53]. Differing criteria were used to define treatment success and failure in the
various studies. Most studies defined “failure” by the occurrence
of objective clinical manifestations, but subjective symptoms

were considered evidence of treatment failure in some studies.

The etiology of residual patient complaints after treatment may include an inflammatory response, unrelated to active infection, or alternative disease processes. Failure rates were not considered in the context of background complaints in an otherwise "healthy" population. For example, in a recent random telephone survey collecting self-reported health information, the prevalence of chronic joint symptoms in adults ranged from 12.3% to 22.7% [54]. In a study of adult members in a health maintenance organization in Seattle, »20% reported fatigue of at least 6 months' duration that interfered with normal activities [55]. Twelve percent of a control group of children without Lyme disease in another study mentioned fatigue as a symptom [56]. In rheumatology practice, a prevalence of 15%–20% for fibromyalgia is common [57]. Nearly 85% of the general population may experience at least 1 somatic symptom in a 6-week period, and 81% of healthy university students and hospital staff members described at least 1 such symptom over a 3-day interval [58, 59]. Thus, the occurrence of arthralgia, myalgia, and fatigue after treatment for early Lyme disease must be

evaluated in the context of background complaints for a significant
proportion of patients.
In addition, the possibility of coinfection with other pathogens
such as *Babesia microti* and the *Ehrlichia* species that causes HGE was not considered in any of the treatment studies
of early Lyme disease. In a separate study in an area in which
babesiosis is endemic, most patients who had residual complaints
after treatment for early Lyme disease had evidence of coinfection with *B. microti* [17]. Specific treatment with antiparasitic
agents directed against this microorganism was effective in diminishing symptoms in 1 study [60].
The first randomized clinical trial of treatment of erythema migrans compared erythromycin, tetracycline, and penicillin at
dosages of 250 mg 4 times daily for 10 days and included 112
adult patients [47]. Signs and symptoms after treatment were
considered to be either “minor” (headache, fatigue, supraventricular
tachycardia, arthralgias, brief arthritis of !2 weeks’ duration, or isolated facial palsy) or “major” (meningitis, meningoencephalitis,
carditis, or recurrent attacks of arthritis).

Approximately 15% of patients had transient intensification of symptoms during the first 24 h of therapy, consistent with a Jarisch-Herxheimer reaction. Erythema migrans and its associated symptoms resolved significantly faster in patients treated with penicillin or tetracycline than in patients treated with erythromycin (*P* ! .05). In addition, treatment with tetracycline or penicillin was associated with a lower rate of occurrence of "major" manifestations by these criteria, compared with the occurrence rate associated with erythromycin.

Overall, "minor" posttreatment signs and symptoms occurred in »45% of patients. Extending therapy to 20 days with tetracycline in a subsequent study by the same investigators had no effect on the frequency of posttreatment symptoms [47].

The results of these studies supported the findings of an earlier open trial of oral penicillin therapy [61]. It could be concluded that erythema migrans was responsive to antibiotic treatment but optimal therapy was not defined.

Subsequent small studies found that doxycycline and amoxicillin (plus probenecid), which are the tetracycline and b-lactam preparations most commonly prescribed in current clinical

practice for patients with erythema migrans, were effective therapies,
and that the efficacies of each drug regimen were similar [48, 49].
A multicenter study that compared cefuroxime axetil (500 mg twice daily for 20 days) with doxycycline (100 mg 3 times
daily for 20 days) in 123 patients with erythema migrans demonstrated
satisfactory outcomes for »90% of patients followed
for 1 year after treatment [50]. Seventy-one percent of patients
in the cefuroxime group and 76% in the doxycycline group were
completely cured, whereas 19% and 16% of patients, respectively,
had persistent subjective complaints but their conditions improved. Although treatment was considered to have failed for 10% of patients, most of these patients did not have objective
evidence of continuing active infection.
A second multicenter study, in which 232 patients with erythema
migrans were randomized to receive either cefuroxime
or doxycycline for 20 days, confirmed that the 2 drugs had comparable efficacy [51]. Consistent with earlier reports, a Jarisch-
Herxheimer–like reaction occurred during the first 24 h of therapy in 12% of patients in each treatment group.
A multicenter, double-blind, randomized prospective trial

compared azithromycin (500 mg once daily for 7 days) with amoxicillin (500 mg 3 times daily for 20 days) in the treatment
of patients with erythema migrans [52]. Amoxicillin was found
to be significantly more effective than azithromycin in resolving
the acute manifestations of erythema migrans completely and
in preventing relapse within a 6-month period. Of 217 evaluable
patients, only 4% of those treated with amoxicillin relapsed,
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Table 3. Recommended antimicrobial regimens for treatment of patients with
Lyme disease.

Recommendation, drug	Dosage for adults	Dosage for children
Preferred oral		
Amoxicillin	500 mg t.i.d.	50 mg/kg/d divided into 3 doses (maximum, 500 mg/dose)
Doxycycline	100 mg b.i.d.	a Age !8 y: not recommended; age >8 y: 1–2 mg/kg b.i.d. (maximum, 100 mg/dose)
Alternative oral		
Cefuroxime axetil	500 mg b.i.d.	30 mg/kg/d divided into 2 doses (maximum, 500 mg/dose)
Preferred parenteral		

Ceftriaxone 2 g iv once daily 75–100 mg/kg iv per day in a single
dose (maximum, 2 g)
Alternative parenteral
Cefotaxime 2 g iv t.i.d. 150–200 mg/kg/d iv divided into 3 or 4 doses (maximum, 6 g/d)
Penicillin G 18–24 million units iv/d
divided into doses
given q4hb
200,000–400,000 units/kg/d, divided
into doses given q4hb (maximum,
18–24 million units/d)
a Tetracyclines are relatively contraindicated for pregnant or lactating women.
b The penicillin dosage should be reduced for patients with impaired renal function.
compared with 16% of those treated with azithromycin (*P*p .005). A higher symptom score before treatment correlated with
persistent symptoms after treatment.
Only 1 study has specifically addressed the treatment of acute
disseminated nonneurological Lyme disease. This prospective,
randomized multicenter trial revealed that in the absence of clinically apparent CNS involvement, oral doxycycline (100 mg
twice daily for 3 weeks) was similar in efficacy to iv ceftriaxone
(2 g daily for 2 weeks) [53].

In most of the controlled trials, patients assigned to be treated with either doxycycline or amoxicillin received therapy for »3 weeks. However, similar success rates have been reported in studies in which 14-day treatment courses with these antibiotics were used [62, 63]. Although none of the prospective studies included pregnant patients, there are no data to suggest that these patients should be treated differently from other patients with Lyme disease, except that tetracycline therapy should be avoided [64].

Several conclusions can be drawn from these trials. Doxycycline, amoxicillin, and cefuroxime axetil are efficacious in the treatment of early Lyme disease. Most patients respond promptly and completely. Some individuals have persistent subjective complaints despite therapy that otherwise appears curative. Less than 10% of infected individuals fail to respond to antibiotic therapy, as evidenced by objective manifestations of persistent infection, and repeat treatment is rarely required. In general, patients who are more systemically ill (e.g., febrile with significant constitutional complaints) at the time of diagnosis

take longer to have a complete response to therapy. Coinfection with other tick-borne infections or inadequately recognized CNS infection at the time of institution of antibiotic therapy may be the explanation for antibiotic failures in some circumstances.

Despite excellent activity against *B. burgdorferi* in vitro [65], the macrolides that have been studied systematically, namely, erythromycin [47] and azithromycin [52] in the United States and roxithromycin [66] in Europe, are less effective than other therapeutic agents (reviewed in [67]). Clarithromycin has not been studied in a controlled trial [68].

All antimicrobials effective in early Lyme disease are associated with a low frequency of serious adverse effects. Druginduced rashes occur with both amoxicillin [52] and cefuroxime [50, 51]. Doxycycline may cause photosensitivity [50, 51], which may be problematic since early Lyme disease occurs most commonly during the summer months. Individuals treated with doxycycline are advised to avoid exposure to the sun while receiving therapy. In addition, doxycycline is relatively contraindicated for children aged !8 years and for women who are pregnant or breast-feeding.

Cefuroxime axetil is much more expensive than doxycycline

or amoxicillin; therefore, its administration is not recommended
as first-line therapy (table 3).
In contrast to the second-generation cephalosporin cefuroxime
and certain third-generation cephalosporins (e.g., ceftriaxone),
first-generation cephalosporins such as cephalexin are inactive
in vitro against *B. burgdorferi* and are ineffective clinically [69, 70].
Available evidence regarding treatment of acute neurological Lyme disease in the United States is based on small case series.
Patients with Lyme meningitis or acute radiculopathy respond
to iv penicillin [71], although ceftriaxone is more widely used for this indication because of its convenient once-daily dosing
[72]. European trials have found iv penicillin to be as effective
as cefotaxime or ceftriaxone [73, 74] and cefotaxime to be as
effective as ceftriaxone [75]. Doxycycline administered orally
or iv has also been used successfully in Europe [76–79], but

Table 4. Recommended therapy for patients with Lyme disease.
Indication Treatment Duration, d

Tick bite None recommended; observe
Erythema migrans Oral regimena,b 14–21
Acute neurological disease
Meningitis or radiculopathy Parenteral regimena,c 14–28
Cranial-nerve palsy Oral regimena 14–21
Cardiac disease
1st or 2d degree heart block Oral regimena 14–21
3d degree heart block Parenteral regimena,d 14–21
Late disease
Arthritis without neurological disease Oral regimena 28
Recurrent arthritis after oral regimen Oral regimena or 28
parenteral regimena 14–28
Persistent arthritis after 2 courses of antibiotics Symptomatic
therapy
CNS or peripheral nervous system disease Parenteral
regimena 14–28
Chronic Lyme disease or post–Lyme disease syndrome
Symptomatic therapye
a See table 3.
b For adult patients who are intolerant of amoxicillin,
doxycycline, and cefuroxime axetil, alternatives
are azithromycin (500 mg orally daily for 7–10 days),
erythromycin (500 mg orally 4 times per day for
14–21 days), or clarithromycin (500 mg orally twice daily for
14–21 days [except during pregnancy]). The
recommended dosages of these agents for children are as
follows: azithromycin, 10 mg/kg daily (maximum,
500 mg/d); erythromycin, 12.5 mg/kg 4 times daily
(maximum, 500 mg/dose); clarithromycin, 7.5 mg/kg

twice daily (maximum, 500 mg/dose). Patients treated with macrolides should be closely followed.
c For nonpregnant adult patients intolerant of both penicillin and cephalosporins, doxycycline (200–400
mg/d orally [or iv if oral medications cannot be taken], divided into 2 doses) may be adequate.
d A temporary pacemaker may be required.
e See the discussion of Chronic Lyme Disease or Post–Lyme Disease Syndrome in the text.
experience with this agent for the treatment of patients with meningitis due to Lyme disease in the United States is limited.
Cranial-nerve palsy has been treated satisfactorily with oral antibiotics [38, 80]. There was disagreement among panel members,
however, on the neurological evaluation of patients with seventh-cranial-nerve palsy. Some members perform a lumbar
puncture for all individuals with Lyme disease–associated seventh-
cranial-nerve palsy. Others reserve lumbar puncture for those patients for whom there is strong clinical evidence ofCNS
involvement (e.g., severe headache or nuchal rigidity). Patients whose CSF examinations yield normal findings may be treated with the same regimens used for patients with erythema
migrans, whereas those with clinical and laboratory evidence of CNS involvement should be treated with regimens effective against meningitis. Since the frequency and rate of

recovery of seventh-cranial-nerve palsy in patients treated with antibiotics appear to be the same as in untreated patients, the principal goal of therapy is to prevent the development of later clinical manifestations [80].

No studies have specifically addressed the treatment of carditis.

Cardiac involvement in North American Lyme disease primarily manifests as atrioventricular heart block and usually occurs within the first several weeks of infection, often in conjunction with erythema migrans [81]. First- and second-degree atrioventricular heart blocks resolve during therapy with oral antibiotics. Because of the potential for life-threatening complications, patients with third-degree atrioventricular heart block should be closely monitored in the hospital. Most panel members treat such patients with iv ceftriaxone, although there is no evidence that parenteral therapy is more effective than oral therapy. Insertion of a temporary pacemaker may be necessary for patients with third-degree heart block in some circumstances.

Recommendations

Administration of doxycycline (100 mg twice daily) or amoxicillin

(500 mg 3 times daily) for 14–21 days is recommended for treatment of early localized or early disseminated Lyme disease associated with erythema migrans, in the absence of neurological involvement or third-degree atrioventricular heart block (tables 3 and 4) (A-I). In prospective studies, these agents have been shown to be effective in the treatment of erythema migrans and associated symptoms.

Doxycycline has the advantage of being efficacious for treatment of HGE, which may occur simultaneously with early Lyme disease. Doxycycline is relatively contraindicated during pregnancy or lactation and for children aged !8 years. Because of its higher cost, cefuroxime axetil, which is as effective as doxycycline in the treatment of erythema migrans (A-I), should be reserved as an alternative agent for those patients who can take neither doxycycline nor amoxicillin. For children, amoxicillin or doxycycline (for those aged >8 years) is recommended (tables 3 and 4) (B-II). Cefuroxime axetil is an acceptable alternative agent (B-III).

Administration of macrolide antibiotics is not recommended as first-line therapy for early Lyme disease (E-I). When used,

they should be reserved for patients who are intolerant of amoxicillin,
doxycycline, and cefuroxime axetil (table 4). Patients treated with macrolides should be closely followed.
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Ceftriaxone (2 g iv daily), although effective, is not superior to oral agents and is therefore not recommended for treatment
of Lyme disease in the absence of neurological involvement or
third-degree atrioventricular heart block (E-I).
The use of ceftriaxone (2 g once daily iv for 14–28 days) in early Lyme disease is recommended for acute neurological disease
manifested by meningitis or radiculopathy (tables 3 and 4) (B-II). Parenteral therapy with penicillin G or cefotaxime may be a satisfactory alternative (B-II). For adult patients who
are intolerant of both penicillin and cephalosporins, doxycycline
(200–400 mg/d in 2 divided doses orally [or iv if the patient is unable to take oral medications]) for 14–28 days may be adequate (B-II).
For children, iv ceftriaxone (B-II) or cefotaxime (B-III) is recommended (tables 3 and 4); penicillin G given iv is an alternative
(B-II).
Patients with first- or second-degree atrioventricular heart block associated with early Lyme disease should be treated in

the same manner as patients with erythema migrans without carditis (tables 3 and 4) (B-III). We recommend that patients with third-degree atrioventricular heart block be treated with parenteral antibiotics such as ceftriaxone (table 3) in the hospital,
although there are no clinical trial data to support this recommendation (B-III). A temporary pacemaker may also be required.
Although antibiotic treatment does not hasten the resolution of seventh-cranial-nerve palsy associated with *B. burgdorferi* infection, antibiotics should be given to prevent further sequelae
(B-II). There was disagreement among panel members on the
neurological evaluation of patients with seventh-cranial-nerve
palsy. Some members perform a CSF examination of all patients
with seventh-cranial-nerve palsy, whereas others reserve lumbar puncture for those in whom there is strong clinical evidence of CNS involvement (e.g., severe headache or nuchal
rigidity).
Patients whose CSF examinations yield normal findings may be treated with the same regimens used for patients with erythema
migrans (B-III). Those with clinical and laboratory evidence of CNS involvement should be treated with regimens effective against meningitis (tables 3 and 4) (B-II).
Treatment for pregnant patients can be identical to that for

nonpregnant patients with the same disease manifestation, except
that tetracyclines should be avoided (B-III).

Late Lyme Disease

Options

The panel considered various oral and parenteral antimicrobial
regimens for treatment of the late manifestations of Lyme disease. Late manifestations include arthritis (oligoarticular), encephalopathy (characterized primarily by memory deficit, irritability,
and somnolence), and neuropathy (manifested primarily by distal paresthesias or radicular pain). Acrodermatitis chronica atrophicans was not addressed because of its rarity in
North America (its primary causative organism, *B. afzelii,* is an exclusively Eurasian genospecies). Because of the lack of evaluable data on ophthalmologic complications, which are very rare, the panel was unable to make recommendations concerning
keratitis and other possible ocular manifestations of Lyme disease.

The response to treatment of late manifestations is typically slow, and improvement or resolution of symptoms may take weeks or months. However, appropriate antibiotic treatment results in eventual recovery in most patients.

Outcome

The panel compared the risks and consequences of ineffective

treatment of late Lyme disease with the problems resulting from adverse effects of antimicrobial therapies. The desired outcome is to treat effectively the late complications of Lyme disease while minimizing the adverse effects of antibiotic therapy. It has not been shown nor is it anticipated that *B. burgdorferi* will develop resistance to antibiotics, but the indiscriminate use of antibiotics exacerbates the problem of antibiotic-resistant community-acquired infections with other bacteria.

Evidence

The first study of antibiotic treatment in patients with Lyme arthritis was initiated in 1980 [82]. The regimens tested were those used for the treatment of tertiary syphilis, and the study design was a double-blind, placebo-controlled trial. The patients had intermittent or chronic Lyme arthritis primarily affecting the knees, and all patients were subsequently shown to be seropositive for antibodies to *B. burgdorferi*. In the first phase of the study, 40 patients were randomized to receive im benzathine penicillin G (7.2 million units) or placebo. In the second phase, 20 patients were treated with iv penicillin G (20 million

units per day for 10 days). Of the 20 patients who received im benzathine penicillin, 7 (35%) had complete resolution of joint involvement soon after treatment, compared with none of 20 patients who were given placebo (*P* ! 0.02). Of the 20 patients treated the following year with iv penicillin G, 11 (55%) had complete resolution of the arthritis soon after treatment. It was concluded that parenteral penicillin was often effective in the treatment of Lyme arthritis, but a number of patients failed to respond.

Subsequently, a series of studies was begun to test the efficacy of iv ceftriaxone in the treatment of late Lyme disease. In comparison with penicillin, the advantages of ceftriaxone are its excellent CSF penetration and long serum half-life, which permits once-a-day dosing for outpatient management. In 1987, a case series of 7 patients with Lyme arthritis or chronic neuroborreliosis, who were refractory to oral or iv penicillin therapy, were then treated with iv ceftriaxone (2 or 4 g/d for 2 S10 Wormser et al. CID 2000;31 (Suppl 1) weeks) [83]. All 5 patients who had arthritis responded to ceftriaxone

therapy, and for 5 of the 6 patients with limb paresthesias, a reduction in symptoms and improvement of nerveconduction study findings were noted.

In a follow-up study, 23 patients with Lyme arthritis or late neuroborreliosis were randomly assigned to receive penicillin (20 million units per day iv for 10 days) or ceftriaxone (4 g/d iv for 14 days) [84]. Of the 13 patients who received ceftriaxone, none had objective evidence of persistent disease after treatment, although 3 had mild arthralgias and 1 complained of fatigue and memory difficulty. In contrast, 5 of the 10 patients who received iv penicillin continued to have fatigue, memory deficit, or recurrent oligoarthritis. For 4 of these 5 patients, symptoms resolved after repeat treatment with ceftriaxone.

In a subsequent study, 31 patients with Lyme arthritis or chronic neuroborreliosis were randomly assigned to receive 2 or 4 g/d of ceftriaxone for 2 weeks [84]. After treatment, 3 of the 31 patients had persistent encephalopathy, 2 had persistent neuropathy, and 3 had no diminishment of their arthritis. The overall frequency of persistent symptoms among patients was 13%, which was similar in both treatment groups. In an open-label, randomized, multicenter study, 143 evaluable patients with manifestations of late Lyme disease, primarily Lyme arthritis, were

treated with iv ceftriaxone (2 g/d for either 2 or 4 weeks) [85].
In 76% of those treated for 2 weeks and 70% of those treated
for 4 weeks, symptoms resolved after treatment (the *P* value was
not significant). The most common persistent symptoms were
arthralgia, pain, weakness, malaise, and fatigue.
The principal conclusions of these 2 studies were that the efficacy of iv ceftriaxone at a dosage of 2 g/d was equivalent to that at a dosage of 4 g/d, and a 2-week course was as efficacious
as a 4-week course for the treatment of late Lyme disease. However, some patients had persistent symptoms despite ceftriaxone treatment.
At the same time that studies were being carried out to assess
parenteral antibiotic regimens, oral therapy was also found to
be effective in the treatment of patients with Lyme arthritis. In
1983 and 1984, 14 children with Lyme arthritis were treated orally with either phenoxymethyl penicillin or tetracycline for 10–30 days [86]. Thirteen experienced no further attacks of arthritis at follow-up at 4–24 months after treatment, whereas
1 patient’s symptoms did not resolve until after a 10-day course
of iv penicillin.

From 1986 through 1991, 48 adult and pediatric patients with Lyme arthritis were randomly assigned to receive either doxycycline (100 mg orally twice a day) or amoxicillin and probenecid (500 mg of each 4 times a day), in each instance for 30 days [87]. Eighteen of the 20 evaluable patients treated with doxycycline and 16 of the 18 evaluable patients who completed the amoxicillin regimen had resolution of arthritis within 13 months after enrollment in the study. However, neuroborreliosis later developed in 5 patients, 4 of whom were treated with the amoxicillin/probenecid regimen. The concomitant use of probenecid with amoxicillin may be inadvisable, because probenecid may impair penetration of b-lactam antibiotics into brain parenchyma [72, 88].

In retrospect, it was noted that all 5 patients reported subtle distal paresthesias or memory impairment at the time of enrollment.

It was concluded that patients with Lyme arthritis can usually be treated successfully with oral antibiotics, but practitioners must be aware of subtle neurological symptoms that may require treatment with iv antibiotics.

In a cost-effectiveness analysis, iv therapy was found to be no more cost-effective than oral therapy for patients with Lyme

arthritis; iv therapy was more likely to result in serious complications
and was substantially more expensive [89]. Therefore, the authors concluded that oral antibiotics are to be preferred
in the initial treatment of Lyme arthritis in the absence of concomitant
neurological involvement.

Not all patients with Lyme arthritis respond to antibiotic therapy. In 1 treatment trial, none of the 16 patients with Lyme
arthritis who were treated with iv ceftriaxone (2 g daily for 2 weeks) had resolution of arthritis within 3 months after completion
of therapy [87]. That study's enrollment requirement of continuous joint swelling for at least 3 months despite treatment
with other recommended parenteral or oral antibiotic regimens
differed from requirements in previous studies.

These 16 patients were also found to have distinctive immunogenetic
and immune markers, including a high frequency
of human leukocyte antigen–DR4 specificity and of antibody reactivity with OspA of the spirochete. More recent data based
on PCR testing of serial joint fluid samples suggest that arthritis
may persist in a small number of patients despite eradication

of the spirochete [90]. The observation that there are epitopes of OspA that cross-react with human leukocyte function–associated antigen-1 [91] suggests that immune phenomena might explain the persistent joint inflammation in these cases.

Arthroscopic synovectomy has been used successfully in the treatment of patients whose arthritis persists despite antibiotic therapy. Of 20 patients who underwent this procedure for refractory chronic Lyme arthritis of the knee, 16 (80%) had resolution of joint inflammation during the first month following surgery or soon thereafter [92]. The remaining 4 patients (20%) had persistent or recurrent synovitis.

Patients with late Lyme disease associated with prominent neurological features also respond to antibiotic therapy. In trials conducted from 1987 through 1989, 27 adult patients with Lyme encephalopathy, polyneuropathy, or both were treated with iv ceftriaxone (2 g/d for 2 weeks) [93]. In addition to clinical signs and symptoms, outcome measures included CSF analyses and neuropsychological tests of memory. Response to therapy was usually gradual and did not begin until several months after

treatment. When response was measured 6 months after treatment,
17 patients (63%) had uncomplicated improvement, 6 (22%) had improvement but then relapsed, and 4 (15%) had no change in their condition.
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In a subsequent study, the same investigators treated 18 adult
patients with Lyme encephalopathy with iv ceftriaxone (2 g/d for 30 days) [94]. Of the 18 patients, 16 had abnormal verbal or visual memory scores on neuropsychological tests and 16 had CSF abnormalities, most commonly production of intrathecal
antibody to *B. burgdorferi* or an elevated total protein level. As determined 6 months after treatment, 14 (93%) of the
15 patients examined had diminished symptoms, and verbal memory scores for the 15 patients were significantly improved
(P ! .01). The total CSF protein values were significantly less for the 10 patients who had follow-up analyses (P ! .05). At 12–24 months, all patients' conditions were back to normal or
improved (1 of the 18 patients was given repeat treatment after
8 months).
It was concluded that Lyme encephalopathy may be associated
with active infection of the nervous system and that the

infection in most patients can be treated successfully with a 30-day course of iv ceftriaxone. Whether a 30-day course is superior to 14 days of treatment is unclear. Although the data are much more limited, the conditions of children with neurocognitive abnormalities attributed to Lyme disease also appear to improve after 2–4 weeks of iv ceftriaxone [95].

The third-generation cephalosporin cefotaxime has been tested in Europe and has been found to be effective in the treatment of late Lyme disease [96]. Although cefotaxime has to be administered 3–4 times daily (compared with once daily administration of ceftriaxone), it does not cause the biliary complications that have been associated with ceftriaxone therapy [97].

Recommendations

Lyme arthritis. Lyme arthritis can usually be treated successfully with antimicrobial agents administered orally or iv (tables 3 and 4). Administration of doxycycline or amoxicillin, in each instance for 28 days, is recommended for patients without clinical evidence of neurological disease (B-II). For children, doxycycline (for those aged >8 years) or amoxicillin is recommended (tables 3 and 4) (B-II). Oral therapy is easier to

administer than iv antibiotics, is associated with fewer serious complications, and is considerably less expensive. Its disadvantage is that some patients treated with oral agents have subsequently manifested overt neuroborreliosis, which may require iv therapy for successful treatment. Further controlled trials are needed to compare oral with iv therapy.

Neurological evaluation, including lumbar puncture, should be done for patients for whom there is a strong clinical suspicion of neurological involvement. Patients with arthritis and objective evidence of neurological disease should receive parenteral therapy with ceftriaxone (tables 3 and 4) (A-II). Alternative parenteral agents include cefotaxime (B-III) and penicillin G (B-II). The long-acting benzathine preparation of penicillin achieves only low levels in the blood and therefore is not recommended (D-III). For children, ceftriaxone iv (B-III) or cefotaxime (B-III) is recommended (tables 3 and 4); penicillin G administered iv is an alternative (B-III).

For patients who have persistent or recurrent joint swelling after recommended courses of antibiotic therapy, we recommend repeat treatment with another 4-week course of oral antibiotics or with a 2- to 4-week course of ceftriaxone iv (tables 3 and 4) (B-III). Clinicians should consider waiting several

months before initiating repeat treatment with antimicrobial agents, because of the anticipated slow resolution of inflammation after treatment. If patients have persistent arthritis despite 2 courses of oral therapy or 1 course of iv therapy, symptomatic treatment with nonsteroidal anti-inflammatory agents is recommended; intra-articular steroids may also be of benefit (B-III). If persistent synovitis is associated with significant pain or if it limits function, arthroscopic synovectomy can reduce the period of joint inflammation (B-II).

Late neuroborreliosis affecting the CNS or the peripheral nervous system. For patients with late neurological disease affecting the CNS or peripheral nervous system, treatment with ceftriaxone (2 g once a day iv for 2–4 weeks) is recommended (tables 3 and 4) (B-II). Alternative parenteral therapy may include administration of cefotaxime (B-II) or penicillin G (BII). Response to treatment is usually slow and may be incomplete. However, unless relapse is shown by reliable objective measures, repeat treatment is not recommended. For children, treatment with ceftriaxone is recommended (tables 3 and 4) (BII). Cefotaxime or penicillin G administered iv are alternatives

(B-II).

Chronic Lyme Disease or Post–Lyme Disease Syndrome

Following an episode of Lyme disease that is treated appropriately, some persons have a variety of subjective complaints (such as myalgia, arthralgia, or fatigue). Some of these patients have been classified as having “chronic Lyme disease” or “post–Lyme disease syndrome,” which are poorly defined entities.

These patients appear to be a heterogeneous group. Although European patients rarely have been reported to have residual infection (or perhaps reinfection) with *B. burgdorferi* [98], this has yet to be substantiated either in a large series of appropriately treated European patients or in a study of North American patients. Residual subjective symptoms that last weeks or months also may persist after other medical diseases (both infectious and noninfectious). It has also been recognized that the prevalence of fatigue and/or arthralgias in the general population is 110% [52–56, 58, 59, 99].

In areas of endemicity, coinfection with *B. microti* or the *Ehrlichia* species that causes HGE may explain persistent symptoms

for a small number of these patients [17, 19]. Randomized controlled studies of treatment of patients who remain unwell
after standard courses of antibiotic therapy for Lyme disease are in progress. To date, there are no convincing published data
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showing that repeated or prolonged courses of oral or iv antimicrobial
therapy are effective for such patients. The consensus
of the IDSA expert-panel members is that there is insufficient evidence to regard “chronic Lyme disease” as a separate diagnostic entity.

Acknowledgments

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References 1-99 Not Listed….. IDSA Treatment Guidelines

Sign_____ Date_____

Print Name_____

INFORMED LYME CONSENT INCLUDES READING THE PRESIDENT OF THE NEW INFECTION SOCIETY OF AMERICA PRESIDENT WHO BELIEVES THE BEST LYME RESEARCH SHOWS:

Lyme is cured in 14-30 Days of antibiotics.

Most Infection Specialists feel Lyme is over diagnosed and is cured in 14-30 Days.

The Use of Antibiotics over 30 days is dangerous and makes super resistant bacteria.

He does not mention co-infections so my impression is they are not a significant or common issue.

"Chronic" Lyme really does not exist at all.

ILADS members treat longer with antibiotics just to get rich.

Sign_____
Date_____Print_____

Bio-Identical Hormone Informed Consent

Women and Men

Currently most physicians feel women should use various synthetic hormones such as synthetic estrogens, horse estrogens, and occasionally low dose synthetic testosterone for perimenopause, menopause and a condition some physicians call "andropause" in men. Others suggest Calcium supplements and plant estrogens.

Most physicians do **NOT** suggest using natural progesterone in any form or dose.

Most physicians do **NOT** support the use of hormones like progesterone for PMS/PDD, and many prescribe antidepressants instead.

Most urologists do **NOT** support the use of any form of testosterone, and if they do it is in the form of low dose patches or gels. Some actually think prostate cancer is caused by testosterone. A small minority of physicians believe that testosterone helps prevent prostate cancer, and that some metabolites of fat estrogen might be more of a promoter of prostate cancer. However, all agree that once you have *any* prostate cancer, *you can never take natural testosterone* because it will usually have receptors for testosterone, and it could easily be made worse with testosterone at that point.

Most physicians do not prescribe bio-identical hormones for post-menopausal women routinely as part of disease prevention, e.g., to prevent thinning bones. Indeed, many prefer the use of bone medications that are reported to prevent fractures. Dr. Schaller defers such decisions on these bone medications to your GYN physician.

Dr. Schaller's father was an OB/GYN physician who had a large number of sisters and daughters. He found benefits with natural progesterone in the 1960's to the 1990's. Yet while he felt his patients did well with natural

progesterone, he felt little research money was spent on studying natural progesterone and other natural hormones -- perhaps because natural hormones are not easy to patent and own. Prometrium is the only natural progesterone hormone replacement that is FDA approved. Dr. Schaller can discuss his experience with its potency during your sessions.

In recent decades, many best selling books have appeared which discuss the issue of aging and hormones. Despite these lay books, most physicians tend NOT to recommend replacement of your bio-identical hormones.

Dr. Schaller is in the MINORITY and feels that since people now may live to be over 95, in stark contrast to 100 years ago, he feels keeping some key basic hormones in the 35 y/o range might be helpful to prevent medical problems. **Yet he offers no promises for this treatment**, and while he feels some studies over the years support his opinion, he is troubled that very few studies on aging look at this issue. He feels it is a neglected topic, or the studies done are not in tune with dosing and practices of some physicians using these hormones.

Most geriatric physicians, internists, urologists and Gynecologists would NOT use progesterone or a little natural testosterone to increase falling levels. They would say this is NOT the standard of medical care, and would suggest that you should age "naturally." Dr. Schaller feels they are sincere in their warnings, and indeed he has seen some anti-aging physicians over-sell natural hormones.

Yet Dr. Schaller's family does use them and he supports their use, but that is really not important, because it is *your opinion* of the evidence and your philosophy that matters. Please ask your other doctors what they think should be done about hormones that are drifting down and then reflect on their advice. Dr. Schaller is not interested in debating this

topic or winning you to his position. **He feels the research is fair over the last 50 years, and he thinks this is partly a philosophical issue. And some studies are interpreted differently based on one's philosophy of aging.**

Further, if you ever do take natural hormones, in which your blood levels are kept in the top half of their *normal* accepted levels, you agree to have a prostate exam 2x per year, and a breast/GYN exam 2x per year. While we respect physicians who feel 1x a year is frequent enough, we have had patient have cancer's found very early and then have a good outcome *due to more regular exams.*

If you are a man, we believe in PSA's done 2x a year, even if you are healthy with no symptoms. This is NOT the standard of care. But again, we have found some very early prostate cancer in new patients, and do not like to see anyone diagnosed seasons after their cancer has started.

I UNDERSTAND MOST PHYSICIANS AND MEDICAL SOCIETIES DO **NOT** SUPPORT DR. SCHALLER'S HORMONE REPLACEMENT APPROACH. **Initials**_____

I UNDERSTAND I DO NOT HAVE TO TAKE DR. SCHALLER'S HORMONES. **Initials**_____

I UNDERSTAND I MIGHT BE OFFERED BIO-IDENTICAL THYRIOD IF MY THYRIOD LEVELS ARE LOW AND I HAVE SYMPTOMS OF DEPRESSION OR FATIGUE, SLOWED COGNITION OR OTHER REASONS. **Initials** _____

I REJECT ANY INTERFERENCE BY MY SPOUSE, CHILDREN, RELATIVES, FRIENDS OR ANY GOVERNMENT AGENCY ON THIS

MATTER. I UNDERSTAND SOME SPECIALISTS CONSIDER SUCH HORMONE TREATMENT MALPRACTICE. NEVERTHELESS, I WISH TO HAVE NATURAL REPLACEMENT HORMONES. **Initials_____**

I UNDERSTAND THAT WE DO NOT HAVE SUFFICIENT RESEARCH ON THE USE OF BIO-IDENTICAL REPLACEMENT HORMONES. **Initials_____**

Sign _____Date_____
Print name_____

IN SIGNING I AGREE TO READ THIS *BEFORE* TAKING ANY NATURAL HORMONE. IF I DO NOT HAVE TIME TO READ IT FULLY DURING MY SESSION WITH DR. SCHALLER, I WILL READ IT BEFORE TAKING THE NATURAL HORMONE (S).

Dr. Schaller is a Mere Consultant:

Informed Consent for a Family Doctor

In Dr. Schaller's intake packet he explains that he will be unable to perform routine physical exams. Primarily because most doctors can do them far, far better. More importantly, while Dr. Schaller has areas of specific interest in his medical practice, he has **not** been trained to think as globally as a primary care physician, such as an internist or

a family doctor. For example, symptoms such as fatigue, trouble concentrating, depression, hormonal changes and body aches can be caused by illnesses which Dr. Schaller is interested in, but can also be caused by **a thousand other illnesses** which a family doctor or internist is trained to evaluate.

Therefore, we require individual's to have a primary care physician who will do their physical exams, and also look for all the possible causes of any complaints you have brought to Dr. Schaller. It is also possible that any problems you bring to Dr. Schaller may have **more than one cause**, and Dr. Schaller might find one of these causes. Yet your family doctor or internist might determine yet **another** important cause.

Even if you have been frustrated in the past with an internist or family doctor, please do not forget they are experts at physical exams, and experts in areas in which Dr. Schaller is untrained. If you do **not** have a family doctor or internist, we request that you see one in the next month. One way to find them is to call your county medical society or to ask local friends what family doctor they use.

Sometimes patients try to have Dr. Schaller replace their family doctor or internist, and this converts Dr. Schaller from a consultant to a primary care physician. He is **not** a primary care physician. **It is not Dr. Schaller's responsibility to repeatedly nag you to have a family doctor or internist. We cannot offer a "nag service**."

By signing this below, you agree to be fully responsible to have an internist or family physician who is involved in your care.

Finally, you are invited to fax or mail a copy of any of your labs or "to do" notes to your family doctor. It is quite possible that at some point on some issues your family doctor and Dr. Schaller may not agree. There are many options to address this problem if it occurs, but it is not unusual for physicians to disagree on some areas of medical care.

Name/Please print _____

Signature _____

Date_____

Mold Illness Treatment Informed Consent

You are being given this form because you might have reported a history of indoor mold exposure or have labs that a very small number of physicians think MIGHT be associated with exposure to mold chemicals on mold spores.

Please understand the following about the Very Controversial Area of Mold Illness:

1) **Most physicians believe mold biotoxins have NO impact on the body other than allergies and asthma.**

2) **Dr. Schaller has no desire to convince you that you should see him instead of an allergist, ENT physician or pulmonologist for your mold or indoor water related illnesses.**

3) **Please see another physician for this type of care if you want a second or third opinion. Dr. Schaller has the names of physicians who believe that indoor mold is generally ONLY a source of allergies, if you want a different type of perspective. We strongly support other input.**

4) **Some government agencies and their scientists are annoyed and *fully oppose* the notion of "sick buildings" or "sick schools."**

5) **Some physicians believe the idea of indoor mold causing illness in people with a normal immune system is Quackery and non-scientific.**

Examples of this position include:

Dr. Gots, MD, PhD who reports:

There is more fear than data concerning the health effects of indoor-air contaminants. Beliefs outpace data. Fear of invisible dangers tends to grow even if confirmatory data is lacking or the fear has been refuted (as by measurements of contaminants).

The symptoms that bring indoor air to the attention of building managers are generally common and nonspecific: fatigue, headaches, and eye and nose irritation. Because almost anything can cause these symptoms, they are not tip-offs of SBS. And patients' belief that a particular building is the culprit can impede medical investigation.

http://web.archive.org/web/20010417165507/www.ictm.com/articles/ICTM119.html

Dr. Paul Lees-Haley also feels mold illness is often nonsense. See this sample excerpt:

Paul R. Lees-Haley, Ph.D., ABPP
Mold Neurotoxicity:
Validity, Reliability and Baloney

The mold neurotoxicity debate is not simply about health care and science -- a focus on money and litigation is pervasive in the communications of the toxic mold promoters. As noted by Paul Scroggins, an environmental engineer with the U.S. Environmental Protection Agency "toxic mold" is a term that "sells." (Scroggins, 2002) The campaign being waged to convince people of the dangers of "toxic mold" is not merely an amusing example of folly in modern

society. The people who are bypassing scientific evidence and engaging in wholesale dissemination of "toxic mold" rhetoric are not neutral forces. If it turns out that these exposures are neuropsychologically harmless, the hysterical claims and unfounded alarms sounded by lawyers, doctors and others will nonetheless have harmed many victims. Who will be responsible for their pain and suffering or emotional distress at being told they are going to die or be permanently brain damaged? On the other hand if we discover evidence of causation of neuropsychological deficits in this area, the findings should be presented widely in a style most helpful to affected patients, not in sensationalized hyperbole. Further exploration of the effects of inhaling mycotoxins and mold spores should be through high-quality, well controlled, scientific studies, not speculation in adversarial settings.

http://www.quackwatch.org/01QuackeryRelatedTopics/toxicmold.html

*****An interesting paper from a lawyer also argues against mold illness:**

Dr. Stephen Redd of the Centers for Disease Control highlighted the difference between ingestion and inhalation and the lack of evidence on the effects of the latter in his testimony before a Congressional Subcommittee on Oversight and Investigations and Housing and Community Opportunity. In his testimony, Dr. Redd noted that the only evidence of tumors being connected to mycotoxin exposure was as a result of chronic ingestion of toxin contaminated foods. While acknowledging the reactive airway and asthmatic effects of inhalation exposure to mold, he went on to state that *there is simply an absence of evidence linking inhalation exposure with pulmonary hemorrhage, memory loss or*

lethargy. **The epidemiological studies of the effects of ingestion exposure should not, on their own, suffice to show consistency of association in a case of inhaled or other type of exposure. Only when a study shows consistency of association in the same manner of exposure will this aspect of the epidemiological criteria be satisfied.**

http://www.rivkinradler.com/rivkinradler/Publications/newformat/200404strober.shtml

Simply, it is important that you fully realize that Dr. Schaller *never* makes any claim to be an expert in this complex mold issue, but he is simply interested in the topic. His comments about mold are *NOT* supported by the CDC, FDA or any medical society or ANY medical specialty board.

He is merely treating you because you request his care and have no other options.

If you feel you have other medical options for this treatment, we encourage you to pursue them, since Dr. Schaller is NOT interested in having many patients reporting illness due to "mold." He does not want to win you to his position.

I HAVE READ THE MATERIAL ABOVE OR I WILL DO SO FULLY IN THE NEXT 24 HOURS.

I UNDERSTAND THE TREATMENTS SUGGESTED BY DR. SCHALLER FOR MY MOLD COMPLIANTS ARE *NOT* THE STANDARD OF CARE IN THE USA OR IN ANY MEDICAL SOCIETY.

NEVERTHELESS, I WANT HIS HELP IF IT MIGHT APPLY TO ME, AND NOT ANOTHER PHYSICIAN'S CARE. *I HAVE BEEN OFFERED OTHER REFERRAL OPTIONS AND I REFUSE THEM*. I WANT DR. SCHALLER TO TREAT ME.

Sign_____

Print Name_____Date_____

POWERFUL IMPEACHMENT OF CONSERVATIVE INFECTION SOCIETY "GUIDELINE" PROCESS

News from Attorney General Richard Blumenthal

Thursday, May 1, 2008
Office of the Attorney General
55 Elm Street
Hartford CT 06106

ATTORNEY GENERAL'S INVESTIGATION REVEALS FLAWED LYME DISEASE GUIDELINE PROCESS, IDSA AGREES TO REASSESS GUIDELINES, INSTALL

INDEPENDENT ARBITER

Arbiter General Richard Blumenthal today announced that his antitrust Investigation has uncovered serious flaws in the Infectious Diseases Society of America's (IDSA) process for writing its 2006 Lyme disease guidelines and the IDSA has agreed to reassess them with the assistance of an outside arbiter.

The IDSA guidelines have sweeping and significant impacts on Lyme disease medical care. They are commonly applied by Insurance companies in restricting coverage for long-term antibiotic treatment or other medical care and also strongly influence physician treatment decisions.

Insurance companies have denied coverage for long-term antibiotic treatment relying on these guidelines as justification. The guidelines are also widely cited for conclusions that chronic Lyme disease is nonexistent.

"This agreement vindicates my investigation – finding undisclosed financial interests and forcing a reassessment of IDSA guidelines, "Blumenthal said. "My office uncovered undisclosed financial interests held by several of the most powerful IDSA panelists. The IDSA's guideline panel improperly ignored or minimized consideration of alternative medical opinion and evidence regarding chronic Lyme disease, potentially raising serious questions about whether the recommendations reflected all

relevant science.

"The IDSA's Lyme guideline process lacked important procedural safeguards requiring complete reevaluation of the 2006 Lyme disease guidelines – In effect a comprehensive reassessment through a new panel. The new panel will accept and analyze all evidence, including divergent opinion. An independent neutral ombudsman – expert in medical ethics and conflicts of interest, selected by both the IDSA and my office – will assess the new panel for conflicts of interests and ensure its integrity."

Blumenthal's findings include the following:

- The IDSA failed to conduct a conflicts of interest review for any of the panelists prior to their appointment to the 2006 Lyme disease guideline panel.
- Subsequent disclosures demonstrate that several of the 2006 Lyme disease panelists had conflicts of interest.
- The IDSA failed to follow its own procedures for appointing the 2006 panel chairman and members, enabling the chairman, who held a bias regarding the existence of chronic Lyme, to handpick a likeminded panel without scrutiny by or formal approval of the IDSA's oversight committee;
- The IDSA's 2000 and 2006 Lyme disease panels refused to accept or meaningfully consider information regarding the

existence of chronic Lyme disease, once removing a panelist from the 2000 panel who dissented from the group's position on chronic Lyme disease to achieve "consensus";

- The IDSA blocked appointment of scientists and physicians with divergent views on chronic Lyme who sought to serve on the 2006 guidelines panel by informing them that the panel was fully staffed, even though it was later expanded.
- The IDSA portrayed another medical association's Lyme disease guidelines as corroborating its own when it knew that the two panels shared several authors, including the chairmen of both groups, and were working on guidelines at the same time. In allowing its panelists to serve on both groups at the same time, IDSA violated its own conflicts of interest policy.

IDSA has reached an agreement with Blumenthal's office calling for creation of a review panel to thoroughly scrutinize the 2006 Lyme disease guidelines and update or revise them if necessary. The panel – comprised of individuals without conflicts of interest – will comprehensively review medical and scientific evidence and hold a scientific hearing to provide a forum for additional evidence. It will then determine whether each recommendation in the 2006 Lyme disease guidelines is justified by the evidence or needs revision of updating.

Blumenthal added, "The IDSA's 2006 Lyme disease guideline panel undercut its credibility by allowing individuals with financial interests – in drug companies, Lyme disease diagnostic tests,

patents and consulting arrangements with insurance companies – to exclude divergent medical evidence and opinion. In today's healthcare system, clinical practice guidelines have tremendous influence on the marketing of medical services and products, insurance reimbursements and treatment decisions. As a result medical societies that publish such guidelines have a legal and moral duty to use exacting safeguards and scientific studies.

"Our investigation was always about the IDSA's guidelines process – not the science. IDSA should be recognized for its cooperation and agreement to address the serious concerns raised by my office. Our agreement with IDSA ensures that a new, conflicts-free panel will collect and review all pertinent information, reassess each recommendation and make necessary changes.

"This Action Plan – incorporating a conflicts screen by an independent neutral expert and a public hearing to receive additional evidence – can serve as a model for all medical organizations and societies that publish medical guidelines. This review should strengthen the public's confidence n such critical standards."

THE GUIDELINE REVIEW PROCESS

Under its agreement with the Attorney General's Office, the IDSA will create a review pane of 8 to 12 members, none of whom served on the 2006 IDSA guideline panel. The IDSA must conduct an open application process and consider all applicants.

The agreement calls for the ombudsman selected by Blumenthal's office and the IDSA to ensure that the review panel and its chairperson are free of conflicts of interest.

Blumenthal and IDSA agreed to appoint Dr. Howard A. Brody as the ombudsman. Dr. Brody is a recognized expert and author on medical ethics and conflicts of interest and the director of the Institute for Medical Humanities at the University of Texas Medical Branch. Brody authored the book, *"Hooked: Ethics, the Medical profession and the Pharmaceutical Industry."*

To assure that the review panel obtains divergent information, the panel will conduct an open scientific hearing at which it will hear scientific and medical presentations from interested parties. The agreement requires the hearing to be broadcast live to the public on the Internet via the IDSA's website. The Attorney General's Office, Dr. Brody and the review panel will together finalize the list of presenters at the hearing.

Once it has collected information from its review and open hearing, the panel will assess the information and determine whether the data and evidence supports each of the recommendations in the 2006 Lyme disease guidelines.

The panel will then vote on each recommendation in the IDSA's 2006 Lyme disease guidelines on whether it is supported by the scientific evidence. At least 75 percent of panel members must vote to sustain each recommendation or it will be revised.

Once the panel has acted on each recommendation, it will have three options: make no changes, modify the guidelines in

part or replace them entirely.

The panel's final report will be published on the IDSA's website.

ADDITIONAL FINDINGS OF BLUMENTAL'S INVESTIGATION

IDSA convened panels in 2000 and 2006 to research and publish guidelines for the diagnosis and treatment of Lyme disease. Blumenthal's office found that the IDSA disregarded a 2000 panel member who argued that chronic and persistent Lyme disease exists. The 2000 panel pressured the panelist to confirm toe the group consensus and removed him as an author when he refused.

IDSA sought to portray a second set of Lyme disease guidelines issued by the American Academy of Neurology (AAN) as independently corroborating its findings. In fact, IDSA knew that the two panels shared key members, including the respective panel chairmen and were working on both sets of guidelines at the same time – a violation of IDSA's conflicts of interest policy.

The resulting IDSA and AAN guidelines not only reached the same conclusion regarding the non-existence of chronic Lyme disease, their reasoning at times used strikingly similar language. Both entities, for example, dubbed symptoms persisting after treatment "Post-Lyme Syndrome" and defined it the same way.

When IDSA learned of the improper links between its panel and the AAN's panel, instead of enforcing its conflict of interest policy, it aggressively sought the AAN's

endorsement to “strengthen” its guidelines’ impact. The AAN panel – particularly embers who also served on the IDSA panel – worked equally hard to win AAN’s backing of IDSA’s conclusions.

The two entities sought to portray each other’s guidelines as separate and independent when the facts call into question that contention.

The IDSA subsequently cited AAN’s supposed independent corroboration of its findings as part of its attempts to defeat federal legislation to create a Lyme disease advisory committee and state legislation and supporting antibiotic therapy for chronic Lyme disease.

In a step that the British Medical Journal deemed “unusual,” the IDSA included in its Lyme guidelines a statement calling them “voluntary” with the ultimate determination of their application to be made by the physician in light of each patient’s individual circumstances.” In fact, United Healthcare, Health Net, Blue Cross of California, Kaiser Foundation Health Plan and other insurers have used the guidelines as justification to deny reimbursement for long-term antibiotic treatment.

Blumenthal thanked members of his office who worked on the investigation….